From Newton to Navier-Stokes

Sean He

How to Use

This handout contains exercises with solutions along with the reading material. The exercises develop important techniques used throughout the rest of the handout, so it is important to do the exercises. It is also highly recommended to read the solutions immediately after doing each exercise, not only for a detailed explanation, but also for additional insights important later on.

You should skim through the first four sections of the Math Review to make sure you know the content. The fifth section is good to read at the start just so you can get as much experience seeing index notation as possible, but both the fifth and sixth sections of the Math Review are not really utilized until the Tensors section.

If this is your first time encountering index notation as well as the Kronecker delta/Levi-Civita symbols, it is also highly recommended to go through the first section of the appendix for additional practice. I separated it from *Section 1.5* on index notation because I wanted to reserve the Math Review section as a reference for all prerequisite knowledge.

Here is the recommended roadmap:

Section 1.1-1.5 → *Section 2-5* → *Section 1.6* → *Section 6* → *Section 10.1* → *Section 7-8* → *Section 10.2* → *Section 9*

Acknowledgements

Huge thanks to my fluid dynamics professor Roman Scoccimarro, whose lectures this handout is adapted from. He gave me excellent physical insight into this topic, which has encouraged me to always try to physically understand the math I'm doing.

Also huge thanks to my fluids TA Panos, who gave me alternate perspectives on certain topics, and also helped me tie together physical intuition with mathematical rigor. In particular, his motivation for introducing the stress tensor is so good I had to steal it!

Special thanks to William Wang for providing very thorough feedback on each part of the handout. Also big thanks to Inhoo Chang for providing feedback.

Introduction

The Navier-Stokes equation... a million dollar equation. It governs all of fluid dynamics (or, well most of it), encapsulating it in two vector partial differential equations:

$$\begin{cases} \dfrac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = -\dfrac{\nabla p}{\rho} + \nu \nabla^2 \vec{v} \\ \nabla \cdot \vec{v} = 0 \end{cases}$$

It's quite scary to look at, isn't it? These equations describe a vast range of phenomena and there are many problems in fluid dynamics that still haven't been solved, such as the existence and uniqueness of solutions to these equations, and how these equations describe turbulence.

However, what we can do is understand what these equations say about fluid flow in general by seeing where they come from. In fact, let me make a bold statement:

> The Navier-Stokes equation is no more than a complicated rewriting of $F = ma$, and with a good foundation in vector calculus, some basic mechanics, and a crash course in tensors, you, the reader, can also derive the Navier-Stokes equation from scratch.

The goal of this handout is to introduce fluid dynamics to the reader from a physics perspective, and guide the reader through the derivation of the Navier-Stokes, and touching on a few relevant and interesting concepts along the way. As opposed to the typical textbook which presents information passively, this handout aims to engage the reader actively by explaining the main ideas, and then breaking up the derivation into small bite-sized exercises. Thus, in the end, the reader will have done most of the heavy lifting.

This method of presentation is inspired by a couple of sources. The first is my fluid dynamics course, in which the professor used the problem sets to not only practice solving problems, but also show interesting concepts not mentioned in lecture. This turned out to be much more memorable for me since I had to take a lot of time to work through problem solving techniques I'd never seen before.

Second is Terence Tao's Analysis textbook, which leave many theorems as exercises, forcing the reader to work through them in order to fully understand the content. I hope working through these exercises not only brings you the sense of accomplishment that comes with deriving an important equation, but also lets you appreciate the subtleties within.

Last is the Berkeley Math Tournament's Power Round, which introduces competitors to topics in higher math related fields through bite-sized problems. In the past the topics have included error correcting codes, as well as graphs and generating functions. It's a pity that there are not as many physics competitions, and of the few that exist, even fewer have these kinds of problems. It's probably out of my capacity to organize a physics competition like that, but with this handout, I hope the people who are interested in more advanced physics can at least get a taste of it.

Oh, one last thing. I want to make it clear that this is a physics handout, not a math handout. As such, rigor and formalism are not the focus, but using physical intuition to build up ideas is the main focus.

With that, have fun!

Contents

1 Math Review

I assume the reader is familiar with vector calculus, so only the parts relevant to this handout are reviewed here. This section also includes some notation conventions we use throughout the rest of the handout. It is highly recommended to read the Index Manipulation section.

1.1 Vector Algebra

We will denote the Cartesian unit vectors here by $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. For two vectors $\vec{a} = \langle a_x, a_y, a_z \rangle$ and $\vec{b} = \langle b_x, b_y, b_z \rangle$, the dot product or scalar product $\vec{a} \cdot \vec{b}$ is defined as

$$\vec{a} \cdot \vec{b} \equiv a_x b_x + a_y b_y + a_z b_z$$

If the angle between the two vectors is θ, then we can also write the dot product as

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

Thus, we can geometrically interpret the dot product as telling us how aligned two vectors are; when the dot product is 0, the vectors are orthogonal.

The cross product or vector product $\vec{a} \times \vec{b}$ is defined as

$$\vec{a} \times \vec{b} \equiv \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y)\hat{\mathbf{x}} + (a_z b_x - a_x b_z)\hat{\mathbf{y}} + (a_x b_y - a_y b_x)\hat{\mathbf{z}}$$

We will not need to deal with other curvilinear coordinate systems so this is enough.

1.2 Derivatives

In fluid dynamics, we will typically be working with at least two spatial dimensions, if not three, so we will almost always deal with multivariable functions. For a function $f(x, y, z)$ we can write its partial derivative with respect to x in the following equivalent ways:

$$\frac{\partial f}{\partial x} = \partial_x f = \nabla_x f$$

The partial derivative tells us how the function changes when we change one of the variables while keeping the others constant. However, the ordinary derivative still retains its usefulness. Consider the case where we want track the value of f for a specific particle, whose position changes as a function of time. That is, let $x = x(t), y = y(t), z = z(t)$. Then, by chain rule,

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}$$

We can also do linear approximations using a first-order Taylor expansion. We can look at the change δf when we change the variables x, y, z by $\delta x, \delta y$, and δz:

$$\delta f \approx \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial z}\delta z$$

Since δx is movement in the x direction, it contributes to δf by a factor of $\partial f / \partial x$, and similarly for the other variables. Of course, if x, y, z themselves depended on t, and we wanted to find δf in terms of δt, we can just use chain rule and say $\delta x \approx (dx/dt)\,\delta t$.

1.3 Vector Calculus Operators

We define the following useful operator, called del:

$$\nabla \equiv \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}$$

For a scalar function $f(x, y, z)$, we define the gradient of f:

$$\nabla f \equiv \frac{\partial f}{\partial x}\hat{\mathbf{x}} + \frac{\partial f}{\partial y}\hat{\mathbf{y}} + \frac{\partial f}{\partial z}\hat{\mathbf{z}}$$

Note that ∇f is a vector field. It has a few useful interpretations. The direction of the gradient tells us the direction in which f is increasing the fastest, and the magnitude tells us how fast it is increasing in that direction. With that, we can rewrite the chain rule in a more compact and informative way. If we let the position vector be $\vec{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$, we have

$$\begin{aligned}
\frac{df}{dt} &= \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} \\
&= \left(\frac{\partial f}{\partial x}\hat{\mathbf{x}} + \frac{\partial f}{\partial y}\hat{\mathbf{y}} + \frac{\partial f}{\partial z}\hat{\mathbf{z}}\right) \cdot \left(\frac{dx}{dt}\hat{\mathbf{x}} + \frac{dy}{dt}\hat{\mathbf{y}} + \frac{dz}{dt}\hat{\mathbf{z}}\right) \\
&= \nabla f \cdot \frac{d\vec{r}}{dt} \\
&= \vec{v} \cdot \nabla f
\end{aligned}$$

This means how fast f changes is related to the spatial rate of change of f (given by ∇f) multiplied by the speed of the particle $\vec{v}$.

We can also define a few more useful operations using del. Given a vector field $\vec{F} = F_x\hat{\mathbf{x}} + F_y\hat{\mathbf{y}} + F_z\hat{\mathbf{z}}$, we define its divergence to be

$$\nabla \cdot \vec{F} \equiv \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

The divergence is a scalar quantity, and it measures how much the vector field is going into or coming out of a point. In terms of fluids, it tells how much a small element of fluid is expanding or shrinking (we will show this later).

The curl is also a useful quantity to define. Given the same vector field $\vec{F}$, we define the curl as

$$\nabla \times \vec{F} \equiv \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial_x & \partial_y & \partial_z \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\hat{\mathbf{x}} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\hat{\mathbf{y}} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{\mathbf{z}}$$

In general, the curl measures how much a vector field is spinning, but in fluids the curl of the velocity is field is known as the vorticity (which is a very important quantity, but we will not use it too much) which measures the rotation of fluid elements.

The final operation we define is the Laplacian of a scalar field f, given by

$$\nabla^2 f \equiv \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

In fact, we can, and we will define the Laplacian for a vector field by simply taking the Laplacian of each scalar component:

$$\nabla^2 \vec{F} \equiv (\nabla^2 F_x)\hat{\mathbf{x}} + (\nabla^2 F_y)\hat{\mathbf{y}} + (\nabla^2 F_z)\hat{\mathbf{z}}$$

1.4 Fundamental Theorems

The three fundamental theorems for multivariable calculus are the Fundamental Theorem of Calculus for Line Integrals, the Divergence Theorem (Gauss' Theorem), and Stokes' Theorem (the Curl Theorem). We will only explain the divergence theorem, since it is the only one we need here.

The divergence theorem states that for a closed surface S and its enclosed region V, for any vector field $\vec{F}$ defined everywhere in V we have

$$\oint_S \vec{F} \cdot d\vec{S} = \int_V \nabla \cdot \vec{F}\, dV$$

Here we omit the double and triple integrals because we are not calculus 3 students and it is easier to write. The left hand side is called the flux because it measures the rate the field lines are leaving the surface. The right hand side essentially sums up all the sources and sinks in the enclosed region (remember the interpretation of divergence). The divergence theorem then tells us these two things must be equal.

In terms of fluids, we can think of the divergence theorem as saying that any net influx or efflux of fluid through a closed surface must be either due to the fluid shrinking or expanding, or the presence of sinks or sources. If divergence is 0 everywhere, then whatever goes in should come out.

1.5 Index Manipulation

When we want to manipulate components of vectors, and later on tensors, it is useful to develop easy to read notation. First of all, instead of using x, y, z for our coordinates, we will use an index: x_1, x_2, x_3. This way, instead of listing out each coordinate individually, we can just write x_i and sum over i.

The other notation shortcut we will use is the Einstein summation convention. When we have indices which are repeated twice, it is implied that they are summed over. For example, we can rewrite the del operator as

$$\nabla = \hat{\mathbf{x}}_i \nabla_i = \sum_{i=1}^{3} \hat{\mathbf{x}}_i \nabla_i$$

where $\nabla_1 = \nabla_x$, etc.

If an index appears more than twice, then usually something has gone wrong, and you have used the same index for two different things, meaning there is ambiguity. If an index appears only once, it is not summed over, but is understood to mean "for any value of that index". It is easiest to understand by seeing some examples.

1.5.1 Kronecker Delta

Suppose we have two vectors $\vec{a} = a_i \hat{\mathbf{x}}_i$ and $\vec{b} = b_j \hat{\mathbf{x}}_j$. When we take the dot product of these two vectors, we want to multiply corresponding components. That is, we want all the products $a_i b_j$ where $i = j$. This motivates the definition for the **Kronecker delta** symbol:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Thus we can write the dot product as

$$\vec{a} \cdot \vec{b} = \delta_{ij} a_i b_j$$

where i and j are summed over. Since the Kronecker delta is only nonzero when $i = j$, we can "set" $i = j$ so that

$$\delta_{ij} a_i b_j = a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3$$

We can think of the Kronecker delta as setting two indices equal to each other, effectively getting rid one of the indices. In fact, we generally skip the step of writing the dot product using the Kronecker delta and directly just use one index. For example, given a vector field $\vec{F} = F_i \hat{\mathbf{x}}_i$, we can write the divergence as

$$\nabla \cdot \vec{F} = \nabla_i F_i = \nabla_x F_x + \nabla_y F_y + \nabla_z F_z$$

1.5.2 Levi-Civita

The other operation we want to be able to compute using index notation is the cross product. Given $\vec{a} \times \vec{b} = \vec{v}$, we would like to define the **Levi-Civita** symbol ϵ such that

$$v_i = \epsilon_{ijk} a_j b_k$$

In other words, summing the RHS over j and k should give us the ith component of $\vec{v}$. We won't need to explicitly manipulate the Levi-Civita symbol, but still, as a reference, it is defined as so:

$$\epsilon_{ijk} = \begin{cases} 0 & \text{if there is a repeated index} \\ 1 & \text{if } ijk \text{ is a cyclic permutation of 123} \\ -1 & \text{if } ijk \text{ is not a cyclic permutation of 123} \end{cases}$$

For example, $\epsilon_{122} = 0$, and $\epsilon_{231} = 1$, since 231 can be obtained by "cycling" 123, but $\epsilon_{213} = -1$ since 213 cannot be obtained by "cycling" 123 (you have to swap two indices at some point). Using these rules, we get

$$\begin{cases} v_1 = a_2 b_3 - a_3 b_2 \\ v_2 = a_3 b_1 - a_1 b_3 \\ v_3 = a_1 b_2 - a_2 b_1 \end{cases}$$

which agrees with the cross product.

One important property we will use is that swapping two indices causes the sign to reverse. For example, $\epsilon_{123} = -\epsilon_{213}$. Or more generally, $\epsilon_{ijk} = -\epsilon_{jik}$ or swapping any two indices.

However, using these rules to compute things with the Levi-Civita symbols is usually very tedious. Thankfully, there is an identity that lets us reduce double cross products (which have two Levi-Civita symbols multiplied together) to Kronecker deltas:

$$\epsilon_{mij} \epsilon_{mkl} = \delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}$$

1.5.3 Caveats

There are a few things to beware of when using index notation. When writing derivatives as ∇_i or ∂_i, it is particularly easy to forget that the derivatives are operators, and not just "multiplied", meaning associativity does not apply. For example, the expressions $k \nabla_i v_i$ and $v_i \nabla_i k$ are not the same.

It is also important to avoid ambiguity. For example, $\nabla_i k v_i$ could mean $\nabla_i (k v_i)$ or $(\nabla_i k) v_i$. This is why we write the unit vectors in front when we write del as $\hat{\mathbf{x}}_i \nabla_i$. For Cartesian coordinates, it does not really make a difference, but taking the derivative of unit vectors in other curvilinear coordinate systems is sometimes nontrivial.

1.6 Linear Algebra Basics

A little linear algebra will be necessary when we introduce tensors.

A **vector space** is a set of objects called **vectors** which are closed under addition and scalar multiplication, meaning, given $\mathbf{v}$ and $\mathbf{w}$ are vectors in the vector space V, we have

$$\mathbf{v}+\mathbf{w} \in V \qquad \text{and} \qquad \alpha\mathbf{v} \in V \text{ for all scalars } \alpha$$

The field (in the algebraic sense) of scalars we consider could depend, but for our purposes, we will always assume $\alpha \in \mathbb{R}$. Also, since we are mainly dealing with 3D space, our vectors will always be $\mathbb{R}^3$ vectors.

A **linear combination** of a list of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is a sum of the form

$$\sum_i \alpha_i \mathbf{v}_1 = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n$$

for scalars $\alpha_i \in \mathbb{R}$.

A **basis** in a vector space V is a set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that for every $\mathbf{v} \in V$, $\mathbf{v}$ admits a unique representation as a linear combination of the basis vectors. For our purposes, since we are always working with $\mathbb{R}^3$ vectors, we will mainly use the standard basis:

$$\hat{\mathbf{x}}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \hat{\mathbf{x}}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \hat{\mathbf{x}}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Given two vector spaces V and W, a mapping between them, $T : V \to W$, is called a linear transformation if the following holds:

$$T(\mathbf{v}_1+\mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2) \qquad \text{and} \qquad T(\alpha\mathbf{v}_1) = \alpha T(\mathbf{v}_1) \text{ for } \alpha \in \mathbb{R}$$

These two properties together are commonly known as linearity, and they allow us to define the transformation by how it acts on a basis in V. Given a basis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in V$, if we know $T(\mathbf{v}_1), T(\mathbf{v}_2), \dots,$ $T(\mathbf{v}_n)$, then linearity tells us we know $T(\mathbf{v})$ for any $\mathbf{v} \in V$.

If V is $\mathbb{R}^n$ and W is $\mathbb{R}^m$, we can also represent T as a matrix by putting $T(\mathbf{v}_i)$ for each i in a row matrix:

$$\begin{bmatrix}T(\mathbf{v}_1) & T(\mathbf{v}_2) & \dots & T(\mathbf{v}_n)\end{bmatrix}$$

We will be mostly working with transformations $T : \mathbb{R}^3 \to \mathbb{R}^3$, so we will be dealing with 3x3 matrices:

$$A = \begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$$

where a_{ij} is the entry in the ith row and jth column.

One operation we will need later is the trace of a matrix. The trace of a matrix is defined for a square matrix, and is given by the sum of the entries on the diagonal:

$$\operatorname{tr} A = a_{11} + a_{22} + a_{33}$$

Since we have Einstein summation notation we can also write it as a_{ii}. As we will see later, the trace of the stress tensor is associated with normal stresses.

2 Properties of Fluids

2.1 What is a Fluid?

You might have learned from chemistry that matter can roughly be separated into three states - solid, liquid, and gas, based on whether they have a fixed shape and volume. Then, we can group liquids and gases together as "fluids". While that is not a bad start, we can be more precise:

Definition A fluid is a substance that deforms continuously under applied shear stress.

What is shear stress? Consider a square. Typically, when we talk about applying a force, we mean applying a force to the center of mass, or equally along a surface. Shear stress arises when we apply varying force along a surface, and two examples are shown below:

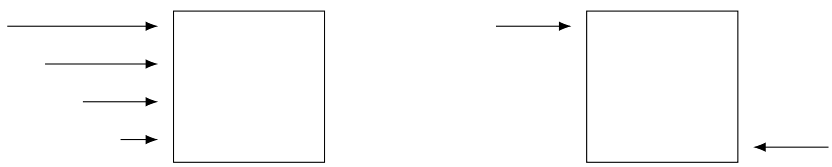


Figure 2.1: Shear stress on square

If the square was some sort of rigid, solid material like wood, applying a small shear force as shown above obviously wouldn't deform it. It would take a significant amount of force to change the shape of the wood square, so in other words, wood does not deform continuously under applied shear stress.

On the other hand, if the square was made of something like water, even the smallest amount of shear force would cause the water to lose its original shape. We might imagine the square as being made up of many layers of water molecules, and when there is a shear force, the layers slide along each other very easily. Thus, water deforms continuously under applied shear stress.

We can then separate fluids into liquids and gases by how they behave under compression. In liquids, the molecules are side by side and hard to compress, while in gases, the molecules are usually much farther apart than their sizes, and thus are easy to compress.

2.2 Continuum Hypothesis

We know that the properties of materials come from molecular interactions at small scales, which is governed by quantum mechanics and statistical mechanics. However, we are interested in studying fluids on scales that are much larger than the molecular mean free path (average distance between collision of molecules). Then, we can ignore quantum effects, and also approximate the fluid to be continuous in space and time.

This assumption that fluids are continuous is what we call the **Continuum Hypothesis**. When we say there is fluid at every point in space at every time, we mean that there is an infinitesimal volume element at the "point" which still contains many molecules. There isn't a good way to mathematically formulate this, but you might think about zooming far enough so that a small volume of fluid looks like a "point". In essence, we want to avoid dealing with quantum and statistical mechanics, but we want to be able to use the nice properties of a continuous fluid and be able to use calculus.

Also, when we use the term "particle", we are referring to a point-mass particle, which will always be a fluid element in our case, and never a single molecule.

3 Fluid Kinematics and Flow Field

Now that we have described the assumptions we are using, and what math we can use to describe fluids, we need to think about how we describe fluid flow. There are two descriptions that naturally arise from what we know.

3.1 Lagrangian Description

Typically, in a first-year mechanics course, we deal with particle mechanics. We describe a particles motion by its position as a function of time, where $\vec{x}(t) = x(t)\hat{\mathbf{x}} + y(t)\hat{\mathbf{y}} + z(t)\hat{\mathbf{z}}$, or whatever coordinate system is convenient. We can do the same in fluid mechanics, except we need to be careful about how we identify "particles" or fluid elements.

Since there is an infinitesimal fluid element at every point at every time, we can identify each one by its position $\vec{x}_0$ at a reference time $t = 0$. Then we can have a displacement field $\vec{\psi}(\vec{x}_0, t)$ which identifies the particle by its initial position and depends on time, allowing us to evolve the particle's position over time. Of course, we want $\vec{\psi}(\vec{x}_0, t = 0) = 0$ so there is no initial displacement.

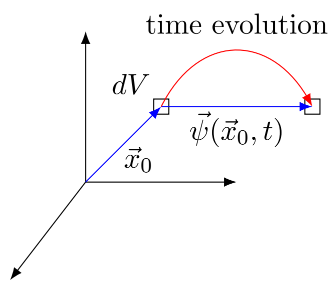


Figure 3.1: Lagrangian fluid element

Exercise 3.1

(a) Using this, write the position of a particle with initial position $\vec{x}_0$ as a function of t.

(b) Differentiate your result from (a) to obtain the Lagrangian velocity $\vec{v}$ of the particles in terms of the displacement field $\vec{\psi}$.

3.2 Eulerian Description

The other natural description of the fluid flow, since we know there is a fluid element at every point, is to take a specific time t and find the velocity of the fluid element at every point. That is, we have a vector function $\vec{v}(\vec{x}, t)$ which gives us the velocity of the fluid element at position $\vec{x}$ and time t.

This is what we call a field description, like the vector fields you encounter in calculus 3. However, note that the while a given fluid element's position may change as a function of time, the velocity field itself can also depend directly on time.

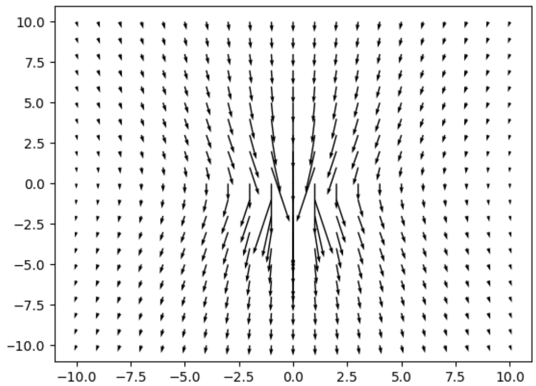


Figure 3.2: Eulerian velocity field

3.3 Relating Both Descriptions

The Lagrangian description is much more intuitive to think about since we follow the same fluid element through its journey. On the other hand, the Eulerian description is much more easy to wield mathematically with all the vector calculus tools that we developed. Thus, it would be useful to relate the two descriptions.

The difference between the two descriptions is that at different times, the Eulerian description may have different fluid elements in the same location, because the fluid elements themselves move, while the Lagrangian description follows that fluid element. Nevertheless, they should describe the same fluid flow.

Exercise 3.2 Consider the path of a fluid element with initial position $\vec{x}_0$ with Lagrangian displacement field $\vec{\psi}(\vec{x}_0, t)$ and Eulerian velocity field $\vec{v}(\vec{x}, t)$. Relate the Lagrangian velocity $\vec{\mathrm{v}}$ and the Eulerian velocity $\vec{v}$. Then write everything in terms of $\vec{\psi}, \vec{x}_0, \vec{v}$ using *Exercise 3.1.*

We will not work too much directly with the kinematics of fluids, but it is important to understand how these two descriptions are related, especially when we study properties other than the velocity.

4 Description of Other Properties and Their Time Evolution

4.1 Material/Total Derivative

Some of the scalar properties of fluid flow that we are interested in include pressure, density, etc. In particular, we are interested what happens over time for a given fluid element.

That is, for a given scalar property $\phi(\vec{x}, t)$ (which we describe as a field), we want to know $d\phi/dt$. This is called the material derivative, or total derivative.

Exercise 4.1 Show that the material derivative is given by

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + (\vec{v} \cdot \nabla)\phi$$

(Hint: Use chain rule.) What do we not account for when we just take a partial derivative w.r.t. time?

The material derivative is also sometimes called Lagrangian derivative because we look at the rate of change of a property as we follow a fluid element, and we can write it in terms of partial derivatives of the property as a scalar field (note that the $\vec{v}$ in the above formula is usually the Eulerian velocity field for this reason).

We can also use the same formula for vector fields, since its the same as taking the material derivative for each component of the vector field. One example which is useful to know is the acceleration, which is the derivative of the velocity.

Exercise 4.2

(a) We first derive the acceleration from the Lagrangian perspective by differentiating the Lagrangian velocity field $\vec{\mathrm{v}}(t)$. (Hint: Rewrite it in terms of the Eulerian velocity field and then use the material derivative)

(b) Now we derive the acceleration from the Eulerian perspective. Consider a velocity field $\vec{v}(\vec{x}, t)$ and a small time perturbation of δt. Note that as time moves forward by $t \mapsto t + \delta t$, the particle's position also evolves by $\vec{x} \mapsto \vec{x} + \vec{v}\,\delta t$. Find how the Eulerian velocity changes by doing a first-order Taylor expansion of

$$\delta \vec{v} \approx \vec{v}[\vec{x} + \vec{v}\,\delta t, t + \delta t] - \vec{v}[\vec{x}, t]$$

(Instead of working with position as a vector, it may be easier to split it up into each component, e.g. $x + \delta x$ where $\delta x = v_x\,\delta t$, and then individually Taylor expand those.)

Using this, find $d\vec{v}/dt$.

Again, since these two descriptions are equivalent, we should get the same result. However, clearly, thinking in the Lagrangian perspective is much easier, and we can use the material derivative to quickly get the answer.

4.2 Physical Interpretation of Divergence

As promised, we will explain why the divergence can interpreted as the expansion/shrinking of a fluid element. First, let us consider a volume of fluid with fixed mass. We can imagine there being a boundary being attached to the fluid, so that the boundary stretches however the fluid elements at the boundary move, so no fluid can escape the boundary. The mass is fixed, but the volume of the fluid can change over time.

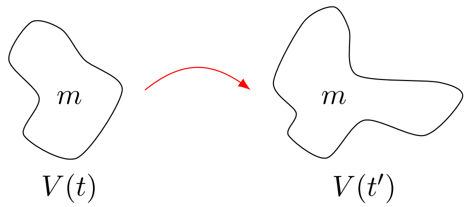


Figure 4.1: Changing volume, fixed mass

There are two ways we can look at how the volume changes over time. One way is to just directly compute the volume, and then take a time derivative. The other way is to look at how the boundary changes. If the boundary stretches, then obviously the volume enclosed increases. To see how the volume changes, let us first consider an infinitesimal fluid element, which we represent as a square, under a velocity field $\vec{v}$:

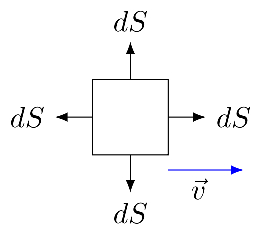


Figure 4.2: Boundary of fluid element under velocity field

The velocity field will push the right surface to the right, increasing the volume at a rate of $v\big|_{\text{right}}\, dS$. Similarly, the left surface will move to the right, causing the volume to decrease at a rate of $v\big|_{\text{left}}\, dS$. If the velocity field is not uniform, then the net change in volume may be nonzero. However, since the velocity is parallel to the top and bottom surfaces, they will only be moved parallel to themselves, and the volume will not change.

Exercise 4.3

(a) We saw that the rate of change of volume depends on the direction of the velocity relative to surface. Given a velocity $\vec{v}$ at a surface element $d\vec{S}$ (magnitude is the infinitesimal area, and direction is the normal to the surface), what should the rate of change of volume be? Use this to write the rate of change of volume for an arbitrary piece of fluid as a surface integral.

(b) Using part (a), show that

$$\frac{d}{dt}\int_V dV = \int_V \nabla \cdot \vec{v}\, dV$$

(c) This gives us a way to relate the divergence to the change in volume for an arbitrary piece of fluid. However, we want to isolate the divergence. Since the above equality is true for any arbitrary volume, we can do this by considering just an infinitesimal volume element δV (this gets rid of the integral). Then, show that

$$\nabla \cdot \vec{v} = \frac{1}{\delta V}\frac{d\delta V}{dt}$$

This technique of converting from the integral form to differential form is very important and will be used over and over.

This says that for an infinitesimal fluid element with fixed mass, the divergence is the rate of change of volume of that fluid element, or its rate of expansion/shrinking. If the mass is not fixed, however, then divergence would indicate there must be fluid entering/exiting somewhere. Since we argued it can't be at the boundary, there must be a source/sink inside the fluid.

4.3 Material Derivative of Unit-Volume and Unit-Mass Quantities

Here we will setup the tools necessary to start deriving our first iteration of the equations of motion for fluids. As explained in the introduction, we will derive them from Newton's 2nd law, which describes particles. This means we will be mainly utilizing the Lagrangian perspective.

Consider a unit-volume quantity ξ. (That is, ξ is something per unit volume, e.g. density is mass per unit volume.) Let's look at an infinitesimal fluid element with volume δV, and track $\xi\delta V$ for the fluid element over time. We don't want to look at ξ itself, because as we just saw, the fluid element can expand or shrink.

Exercise 4.4

(a) Show that

$$\frac{d}{dt}(\xi\,\delta V) = \left(\xi\nabla\cdot\vec{v} + \frac{d\xi}{dt}\right)\delta V$$

(Hint: The result we just derived for the divergence will be useful.)

(b) Now consider the same thing for an arbitrary finite volume by integrating the equation from part (a) over a volume V. This is also commonly known as **Reynolds Transport Theorem**.

Note: Let's see what happens in the Eulerian case. Because we look at a fixed position, ξ only explicitly depends on time, and the coordinates don't depend on time when integrating over the volume. Thus, we have

$$\frac{\partial}{\partial t}\int_V \xi\,dV = \int_V \frac{\partial\xi}{\partial t}\,dV$$

This is the usual trick of differentiating under the integral sign, but we see it only works because we fixed the position.

Now let's consider a unit-mass quantity φ. Once again, we're interested in $\varphi\rho\,\delta V$, where $\rho\,\delta V$ is the mass of the fluid element.

Exercise 4.5

(a) Show that the material derivative is given by

$$\frac{d}{dt}(\varphi\rho\,\delta V) = \rho\,\delta V\frac{d\varphi}{dt}$$

(Hint: Can you use a physical argument to explain why the other term from product rule must be 0?)

(b) Again, consider a finite volume and derive the integral form of the above equation.

Note: The same remark about the Eulerian case applies here too!

Now we are ready to derive our equations of motion.

5 Ideal Equations of Motion

5.1 Continuity Equation

In particle mechanics, we deal with discrete objects, meaning we usually don't have to worry about changing mass. However, since we are dealing with a continuum in fluid mechanics, we need to impose an extra constraint for mass conservation. As we follow the motion of a fluid over time, we want the material derivative $dm/dt = 0$.

Exercise 5.1

(a) Write the mass m of a portion of fluid as the integral of a unit-volume quantity.

(b) Rewrite mass conservation ($dm/dt = 0$) using (a) and derive the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

where ρ is the density of the fluid (not necessarily constant!).

(Hint: Use *Exercise 4.4*. If you are stuck, try working backwards and expanding $\nabla \cdot (\rho \vec{v})$ by component and then using product rule. If you have not read *Section 1.5*, now is a good time to do so.)

(c) One important case is when the fluid is incompressible, which means the desnity is constant everywhere. What does the continuity equation reduce to in this case?

It turns out the differential form for conservation laws all take this form:

$$\frac{\partial}{\partial t}(\text{density of something}) + \nabla \cdot (\text{flux of something}) = 0$$

The flux is given by the density times the velocity. For example, in electromagnetism, charge conservation is given by the exact same equation, where ρ is the charge density, and $\vec{j} = \rho \vec{v}$ is the current density, which is also the charge flux.

To see where this comes from, let's consider an alternate derivation of the continuity equation using an Eulerian perspective.

Exercise 5.2

(a) Consider a fixed region in space defined by a surface S and enclosed volume V. Recall the argument we used in *Exercise 4.3* for the divergence. Similarly, write the flux of mass as a surface integral. (Note that this flux is nonzero because we fixed the region in space.)

(b) Similarly, show that

$$\frac{\partial}{\partial t} \int_V \rho \, dV + \int_V \nabla \cdot (\rho \vec{v}) \, dV = 0$$

(Pay attention to the sign, since we use the convention of outward normal, but an outward flux of mass is a decrease in mass in the region.)

(c) Finally, convert the equation into differential form to get the continuity equation. (Why can we just put the time derivative inside the integral as a partial derivative?)

In other words, the continuity equation tells us that for a fixed region, the rate of change of mass is equal to the flux of mass out of the region. If this were not true, that would imply there is some sort of source or sink inside the region, meaning mass is not conserved.

5.2 Momentum Equation

As promised, our main equation of motion comes from Newton's 2nd Law, which tells us

$$\vec{F} = \frac{d\vec{p}}{dt}$$

where $\vec{p}$ is the momentum.

Exercise 5.3

(a) Write the momentum $\vec{p}$ as the integral of a unit-mass quantity.

(b) Define $\vec{f}$ as the force per unit volume, so we can also write $\vec{F}$ as a volume integral. Then derive the momentum equation

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = \frac{\vec{f}}{\rho}$$

(Hint: Use *Exercise 4.5*.)

Even in this form, Newton's 2nd Law is still recognizable. We see that the left hand side is just the material derivative of the velocity, or the acceleration, and $\vec{f}/\rho$ is the force per unit mass.

5.3 What is Pressure?

How do we exactly describe the forces acting on our fluid? First of all, there are contact forces and long range forces (due to fields). Long range forces include gravitational forces, electromagnetic forces, etc. Let us first deal with contact forces, which we can split into normal and tangential stresses.

Let's look at the simplest case: equilibrium. In this case, the velocity is 0, and all the forces must be in equilibrium. Furthermore, there cannot be any tangential stresses (like friction) in equilibrium because or else the layers of the fluid would start sliding against each other, making it no longer be in equilibrium (we said fluids deform continuously under shear stress). Thus, we will naturally first consider an ideal fluid without tangential stresses.

So how do we describe the normal stresses in equilibrium? Let's look at this informal geometric argument from Pascal. Consider a triangle in the fluid:

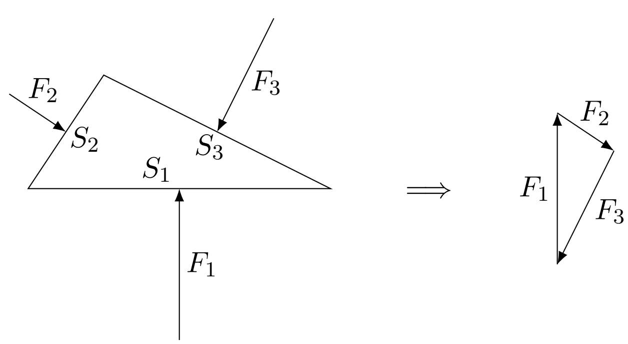


Figure 5.1: Normal forces on triangle in fluid in equilibrium

Since the fluid is in equilibrium, the normal forces should add up to 0, meaning if we add the vectors tip-to-tail, they should form a closed loop. We see that the two triangles should be similar, meaning the ratio between the forces and the surface should be constant. This allows us to define a new quantity called pressure:

$$p \equiv \frac{F_1}{S_1} = \frac{F_2}{S_2} = \frac{F_3}{S_3}$$

In general, we can define a pressure field $p(\vec{x}, t)$ for every point in space and time, which we can use to generalize the relationship between force and pressure to an arbitrary surface. We use the sign convention that positive pressure means compression. Then, for an arbitrary surface element dS, the normal force on it should have magnitude $p\,dS$ and it should point in the normal direction $-\hat{n}$ due to the sign convention. Thus, for an arbitrary surface, we can integrate to get

$$\vec{F} = -\oint_S p\hat{n}\,dS = -\oint_S p\,d\vec{S}$$

5.4 Euler Equation

Now that we can use pressure to describe the forces on the fluid, we can modify our momentum equation to be more specific.

Exercise 5.4

(a) We first need to be able to relate pressure to the unit-volume force $\vec{f}$. To do this, show that

$$\oint_S p\,d\vec{S} = \int_V \nabla p\,dV$$

(Hint: Use the divergence theorem on each component of the gradient. We have $\nabla p = \partial_x p\hat{\mathbf{x}} + \partial_y p\hat{\mathbf{y}} + \partial_z p\hat{\mathbf{z}}$. Can you write $\partial p/\partial x$ as the divergence of something?)

(b) Use this to derive the Euler equation

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = -\frac{\nabla p}{\rho}$$

To sum up, we found that in the static case, we could define a useful quantity called pressure to describe the normal stresses. Then, we used that to modify the momentum equation to describe the normal forces that result from pressure, but we temporarily ignored tangential forces (which come from a property of fluids called viscosity).

If we want to include long-range forces, such as gravity, we can simply add a field acceleration, like $\vec{g}$, to the right hand side of the equation.

Thus, we have derived our equations of motions for an ideal fluid (no tangential stresses):

$$\begin{cases} \dfrac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \\ \dfrac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = -\dfrac{\nabla p}{\rho} \end{cases}$$

You may notice that these equations are not closed since we have five unknowns (ρ, p and three components of $\vec{v}$) and we only have four equations (one from continuity and three from the three components the Euler equation). These equations are all mechanics can tell us about fluids, so to fully solve the equations, we need an additional assumption, like constant ρ, or a relation between p and ρ, which we can obtain from thermodynamics (often the ideal gas law).

These equations are nonlinear partial differential equations, which means they are very difficult to solve generally. We will not discuss solutions to these equations, but if you have solved differential equations, you know that we need boundary conditions in order to find a particular solution.

This makes sense since these equations describe any ideal flow, but obviously there are many different flows. The boundary condition we use is that the fluid does not penetrate any solid boundary. If there is any normal velocity at a physical boundary, then that would mean the fluid is flowing past the boundary, which is not possible. Thus, we should have

$$\vec{v} \cdot \hat{n}\big|_{\text{boundary}} = 0$$

This concludes our discussion of the equations of motion for ideal fluids.

6 Tensors

Now we need to find a way to describe the tangential stresses as well, but it turns out we need to develop some more mathematical machinery.

6.1 What is Pressure . . . ?

In *Section 5.3*, we defined pressure for a fluid in static equilibrium, but we never explained why we expected the same thing to be true for a fluid in motion. In fact, it is not true in general, and we expect that the flow of fluid will influence the normal stresses, and we will see that part of the normal stress depends on velocity.

However, the bigger problem is how we generalized our informal geometrical argument. We defined pressure as the normal force per unit area by considering a polygon, and then generalized it to a continuous scalar field. When we consider the pressure on a specific surface, it is unambiguous what the normal direction in which pressure acts, but in general, what direction should pressure act in?

A common argument you might have heard in high school physics class is that pressure acts in all directions. This actually comes from the continuum hypothesis: Consider a fluid element. No matter which direction we look, there is always another fluid element next to it in that direction, meaning there is a normal contact force acting in that direction.

Thus, it would be more accurate to say that the direction pressure acts in depends on the choice of surface. For example, if we consider the xy plane, the z direction would be the direction pressure acts in, but if we consider the yz plane, no force due to pressure will be present in the z direction. Since we want to consider tangential stresses as well, which are not present in the first case, but are present in the second case, it is even more crucial we distinguish between them.

This suggests we need some new structure to describe stress. What we would like to be able to do is take in information about the orientation of the surface we are looking at (by considering its normal direction), and the direction in which we want to measure stress, and then spit out a number which gives us the force per unit area in that direction.

In mathematical terms, given two vectors $\vec{v}, \vec{w} \in V$ (in this case $\mathbb{R}^3$), one of them being the normal the surface we are considering, and the other being the direction of stress we want to find, we want a function $\sigma : V \times V \to \mathbb{R}$ such that $\sigma(\vec{v}, \vec{w})$ gives us the value of the stress. By examining the properties our function should have, we will see that the function has the structure of what we call a tensor.

6.2 What is a Tensor?

We claim that this function should be linear in each of the vectors it takes in. That is,

$$\sigma(\alpha\vec{v}_1 + \beta\vec{v}_2, \vec{w}) = \alpha\sigma(\vec{v}_1, \vec{w}) + \beta\sigma(\vec{v}_2, \vec{w}) \qquad \text{for } \alpha, \beta \in \mathbb{R}$$

and similarly for $\vec{w}$. Physically, this says that when we add together the directions of stress, we should also add the stress, and similarly for the orientation of the surface. Let us try to provide some physical intuition as to why this might be true.

It is straightforward for the direction of stress. Consider a fixed surface. Then, the net force on that surface is fixed. If we want to find the force in a certain direction, then we simply dot it with the unit vector in that direction, and it is well known that the dot product is linear with respect to each vector it takes in. Physically, this boils down to the fact that we describe forces using vectors, so we can add the stresses as well as long as we add the directions of the stresses like vectors.

It is a little harder to see why stress should be linear with respect to the normal direction of the surface we consider. For the sake of simplicity, we keep the argument informal and avoid heavy mathematical machinery. We consider the relationship of stress to something we know that acts linearly: momentum. Turns out we can write the momentum equation in conservation form like we did with the continuity equation, with a new object called the momentum flux, which we can literally think of as the flux of momentum through a surface. Since momentum is not conserved ($dp/dt = F$), momentum flux is not only due to the fluid flow itself, but also due to stress. In short, Newton's 2nd Law says that stress on a surface causes momentum flux through that surface. We know that momentum flux is linear with respect to the normal of the surface since we just dot the normal with the momentum, and thus, the stress is also linear with respect to the normal of the surface.

Hopefully it is a little clearer now why the stress function we want should be linear in each of its inputs. Such a function is called a multilinear mapping, and since it spits out a scalar, we call it a **multilinear form**, which is exactly what a **tensor** is (or would be, if anyone really knew what a tensor was).

The stress tensor σ that we will examine is a bilinear form, or **rank 2 tensor**, because it takes in two vectors, representing the direction of stress and direction of normal to surface.

6.3 Representation of Tensors

It's a bit unwieldy to have to deal with a transformation that takes in a two arbitrary vectors and gives a scalar, but fortunately, it is precisely the multilinearity of the tensor that saves us.

If you have taken linear algebra, you know that any linear mapping is completely defined by how it acts on a basis. What this means is that if we know how the mapping acts on something like the standard basis $\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \hat{\mathbf{x}}_3$ (which will usually be x, y, z for our purposes), then we know how the mapping acts on any vector $\vec{v}$ since we can write it as a linear combination of the basis vectors: $\vec{v} = v_1\hat{\mathbf{x}}_1 + v_2\hat{\mathbf{x}}_2 + v_3\hat{\mathbf{x}}_3$ for some $v_1, v_2, v_3 \in \mathbb{R}$. For the stress tensor, this means that we just need to know $\sigma(\hat{\mathbf{x}}_i, \vec{w})$ for each i and for a given $\vec{w}$ in order to know $\sigma(\vec{v}, \vec{w})$ for any $\vec{v}$ and a given $\vec{w}$. Then we do the same for $\vec{w}$.

Thus, we just need to know $\sigma(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j)$ for every combination of $i, j = 1, 2, 3$. It's getting cumbersome to write out the unit vectors, so let's use the shorthand $\sigma_{ij} = \sigma(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j)$. This might motivate us to write everything as a matrix:

$$\sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

We know this matrix contains all the information needed to completely define the tensor, but how does it exactly "take in" two vectors and give back a scalar? Let's again think about what we physically expect the stress tensor to do.

We first need to disambiguate which vector is the stress direction and which vector is the normal to the surface. For $\sigma(\vec{v}, \vec{w})$, we usually take the first vector $\vec{v}$ (described by the first index in the matrix) to be the stress direction, and the second vector (described by the second index) to be the normal to the surface.

Exercise 6.1

(a) Physically interpret the meaning of σ_{ij}. Which entries in the matrix are the normal stresses? Which ones are the tangential stresses?

(b) Given vectors $\vec{v} = v_i \hat{\mathbf{x}}_i$ and $\vec{w} = w_j \hat{\mathbf{x}}_j$, show that

$$\sigma(\vec{v}, \vec{w}) = v_i \sigma_{ij} w_j$$

(c) Let's try to understand this in two parts: 1. First, we take in the surface normal $\vec{w}$. 2. Then, we take in the stress direction $\vec{v}$.

   i. Using the matrix we defined above, write $\sigma_{ij} w_j$ as a matrix-vector product. Note that multiplying a vector with a matrix gives another vector. But what does this vector represent?

   ii. To find out, rewrite $\sigma_{ij} w_j$ as $1 \cdot \sigma_{ij} \cdot w_j$ and interpret this physically. How does this relate to the components of the matrix-vector product? Can you interpret the matrix σ as a linear transformation?

   iii. The components of the matrix-vector product are given by $u_i = \sigma_{ij} w_j$, so the stress should be given by $\sigma(\vec{v}, \vec{w}) = v_i u_i$. Rewrite $v_i u_i$ in terms of the vectors $\vec{v}$ and $\vec{u}$, and considering the physical interpretation of $\vec{u}$, explain why $v_i u_i$ should give us the stress in the direction of $\vec{v}$.

To recap, we can represent the stress tensor as a matrix, and we can use vector operations to write the stress as so:

$$\sigma(\vec{v}, \vec{w}) = \vec{v} \cdot (\sigma \vec{w})$$

We first multiply the stress tensor by the normal to the surface, which gives us a stress vector, and then we dot it with the direction we want to measure in to find the stress in that direction.

You might also see this written in a few other ways. Mathematicians will use column vectors (e.g. $\mathbf{v} = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}^T$) and write it as:

$$\mathbf{v}^T \sigma \mathbf{w}$$

Physicists like to view the vectors as being "dotted" into the tensor, so they will write:

$$\vec{v} \cdot \sigma \cdot \vec{w}$$

So $\hat{\mathbf{x}}_i \cdot \sigma \cdot \hat{\mathbf{x}}_j$ can be thought of as "selecting" the i, j entry of σ.

6.4 Isotropic Tensors

One of the key assumptions that will simplify our derivation of the form of the stress tensor is that our fluid is **isotropic**, meaning it has no preferred direction. The stress should of course depend on direction, but only as much as the fluid flow depends on direction. The stress tensor itself should not depend on direction independent of the flow.

A few examples of media that are not isotropic include fluids that are somehow polarized or magnetized, as well as solids that have a certain lattice structure, meaning there are certain preferred directions of movement. (The stress tensor is used throughout continuum mechanics, which describes both continuous solids and fluids.)

For this purpose, we need to define isotropic tensors. The word isotropic comes from the Greek roots iso-, meaning "equal", and -tropic, which means "turn". Thus, we should understand "no preferred direction" as meaning "invariant under rotations".

To try to motivate what it means to rotate a tensor, let's consider an analogy with rotating a vector. When we rotate a vector, we are rotating the components of the vector itself, but we can also think of it as rotating the axes the opposite way:

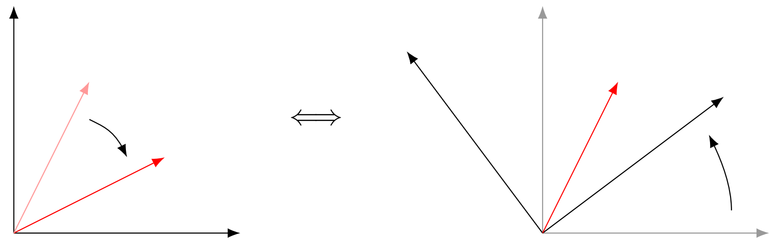


Figure 6.1: Rotating a vector

Similarly, we might think of rotating a tensor as rotating all the vectors it takes in (like rotating the axes), and then defining the composition as the rotated tensor.

For example, consider a tensor T which takes vectors $\vec{u}, \vec{v}, \dots, \vec{w}$ to the scalar $T(\vec{u}, \vec{v}, \dots, \vec{w})$. We can define the rotated tensor by taking the same vectors to $T(R\vec{u}, R\vec{v}, \dots, R\vec{w})$ for a given rotation R. (We first rotate the vectors, and then plug it into the function.)

The question is what something like $T(R\,\cdot\,, R\,\cdot\,, \dots, R\,\cdot\,)$ actually is, and how we should write it down explicitly. Let's write things in index notation. The tensor must have an index for each vector it takes in so we can write it as $T_{pq\dots r}$. The first index must correspond to the rotated vector $R\vec{u}$, so the tensor should be multiplied by $(R\vec{u})_p$. But $R\vec{u}$ is just a matrix vector multiplication, which we can write out in component form as $R_{ip}u_i$. Thus, we must have

$$\begin{aligned} T(R\vec{u}, R\vec{v}, \dots, R\vec{w}) &= T_{pq\dots r}(R\vec{u})_p(R\vec{v})_q \cdots (R\vec{w})_r \\ &= T_{pq\dots r}(R_{ip}u_i)(R_{jq}v_j)\cdots(R_{kr}w_r) \\ &= (R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r})u_i v_j \cdots w_r \end{aligned}$$

It makes sense to define the rotated tensor as

$$A_{ij\dots k} = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r}$$

We can think of each index being "rotated" from $p, q, \dots, r$ to $i, j, \dots, k$. Now if this tensor is invariant under the rotation, it must mean we have $A = T$. This motivates the definition for an isotropic tensor:

Definition A tensor T is isotropic if given any rotation R, the tensor is invariant when we apply the rotation to each of the vectors the tensor takes in.

For any given rotation specified by its matrix $R_{\alpha\beta}$, a tensor $T_{ij\dots k}$ is invariant if

$$T_{ij\dots k} = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r}$$

Turns out we have already seen tensors which are isotropic - the Kronecker delta is an isotropic rank-2 tensor. We didn't introduce the Kronecker delta as a tensor, but it clearly is one: it takes in two vectors and compares components, giving 0 or 1 depending whether the indices of the components are equal. You can also check that multilinearity holds. If we write it as a matrix, we see that $\delta_{ij} = I$, the identity matrix.

In fact, we can show δ_{ij} is the only isotropic rank-2 tensor (up to a constant of proportionality) by considering a few special rotations.

Exercise 6.2 Suppose T_{ij} is a rank-2 isotropic tensor.

(a) Consider a rotation about the x_3(or z)-axis of $\pi/2$. Then the rotation matrix is given by:

$$R = \begin{bmatrix} \cos(\pi/2) & -\sin(\pi/2) & 0 \\ \sin(\pi/2) & \cos(\pi/2) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Since T_{ij} is isotropic, we have

$$T_{ij} = R_{ip}R_{jq}T_{pq}$$

Our goal is to use this to show that $T_{ij} = \alpha\delta_{ij}$ for some α. Let us start with an example:

$$T_{13} = R_{1p}R_{3q}T_{pq}$$

This is summed over p and q, but checking the matrix, there is only one nonzero entry in each of the rows, so only one choice of p and q makes the product nonzero, giving

$$T_{13} = R_{1p}R_{3q}T_{pq} = R_{12}R_{33}T_{23} = -T_{23}$$

i. Similarly, show that $T_{23} = T_{13}$. Then conclude that $T_{13} = T_{23} = 0$.

ii. Similarly, show that $T_{31} = T_{32} = 0$.

iii. Finally, show that $T_{11} = T_{22}$

(b) Now consider a rotation of $\pi/2$ about the x_1-axis. Here is the rotation matrix for your convenience:

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

i. Show that $T_{12} = T_{21} = 0$.

ii. Show that $T_{22} = T_{33}$.

(c) Thus conclude that $T_{ij} = \alpha\delta_{ij}$ for some α.

Hopefully, the concept of isotropic tensors feels a little more concrete now. We will not be repeating these calculations anymore, because they are tedious, and not enlightening. It is enough to know what the isotropic tensors are.

Rank-0 tensors are scalars, which are always invariant under rotations. Rank-1 tensors are vectors, which have a direction, so intuitively there are no isotropic rank-1 tensors.

The only isotropic rank-3 tensor is the Levi-Civita tensor ϵ_{ijk}.

What we are really interested in (a.k.a what will show up later) is the form of isotropic rank-4 tensors. It turns out we can construct them purely out of Kronecker deltas, and any isotropic rank-4 tensor can be written in the form

$$T_{ijkl} = \alpha\delta_{ij}\delta_{kl} + \beta\delta_{ik}\delta_{jl} + \gamma\delta_{il}\delta_{jk}$$

Of course we will not show this, because dealing with 81 coefficients is a monumental pain and can be done by a computer.

Note: We could have also tried to construct it out of Levi-Civita tensors, but we can always reduce it back to Kronecker deltas with the identity

$$\epsilon_{mij}\epsilon_{mkl} = \delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}$$

7 The Stress Tensor

It is highly recommended to work through *Section 10.1* before reading this section.

Now we have two big questions left to answer: 1. What does the stress tensor look like exactly? 2. How do we modify our equations of motion to include the stress tensor?

7.1 What is Viscosity?

Let us answer the first question first. The real answer is that we don't really know what the stress tensor should be. There are a lot of factors that determine whether certain properties of viscosity are negligible or not, and it's hard to figure out exactly when what happens. What we *can* do is reason carefully and clearly lay out our assumptions, which will give us a pretty reasonable answer.

We argued earlier that pressure (or what we should call thermodynamic pressure) is only really accurate for the static equilibrium case. However, this does not mean it is not a useful quantity to define. Thus, we can split the stress tensor into pressure and a viscous part:

$$\sigma_{ij} = -p\delta_{ij} + \sigma'_{ij}$$

Note that the minus sign in front of the pressure is because of our sign convention that positive pressure is compression. What we are left with is the viscous stress tensor σ'_{ij}, which accounts for the effects of viscosity.

But what exactly is viscosity? It's not exactly just the effects of tangential stresses, but rather a general resistance to deformation - basically a fluid analog to friction. Tangential stresses only appear when we include viscosity but because we're working with a continuum, there might also be additional normal stresses (like we predicted) due to a resistance to change in volume. This means the viscous stress tensor should still have nonzero trace.

7.2 Dependence on Physical Properties

In order to figure out what the viscous stress tensor depends on, we need to figure out what it can't depend on.

Consider a uniform flow given by $\vec{v} = u(t)\hat{\mathbf{x}}$. This is a time-dependent flow, but it is still uniform at every instant. This means there is no shear stress (or tangential stress) and the viscous stress tensor should disappear. This means that stress cannot depend directly on velocity or its time derivatives, and should therefore depend on its spatial derivatives. (The same argument tells us that it should depend on derivatives of velocity rather than momentum because density doesn't change in a uniform flow.)

Here we make our first big assumption. We assume that the stress only depends on derivatives of the first order, and also depend on them linearly. Such fluids are called **Newtonian fluids**. This assumption works pretty well as long as the molecular structure of the fluid is negligible. So we say

$$\sigma'_{ij} \propto \frac{\partial v_k}{\partial x_l}$$

We use different indices because the stress could depend on all the different combinations and its derivatives, e.g. we're not sure if σ'_{xy} could dependence on z or z-components like $\partial v_z/\partial x$ or $\partial v_x/\partial z$.

Exercise 7.1 We can use some physical intuition to narrow down which derivatives of which components of velocity we expect the stress to depend on. Recall the physical interpretation of σ_{ij} and what it means to have shear stress. Which derivatives look like they should be responsible for the shear stress? Which derivatives should be responsible for any extra normal stress?

We can also still glean some more information about the general form of the dependence on velocity derivatives. We can always separate the derivatives into a *symmetric* and *antisymmetric* part, like so:

$$\frac{\partial v_k}{\partial x_l} = \left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} + \left.\frac{\partial v_k}{\partial x_l}\right|_{\text{antisym}}$$

where

$$\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} = \frac{1}{2}\left(\frac{\partial v_k}{\partial x_l} + \frac{\partial v_l}{\partial x_k}\right) \qquad \text{and} \qquad \left.\frac{\partial v_k}{\partial x_l}\right|_{\text{antisym}} = \frac{1}{2}\left(\frac{\partial v_k}{\partial x_l} - \frac{\partial v_l}{\partial x_k}\right)$$

This was not split arbitrarily - each part also has a physical interpretation. The symmetric part is known as the rate of strain tensor, which measures the rate of the deformation of the fluid. To get an idea of where this comes from, look at the trace of this tensor:

$$\delta_{kl} \left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} = \frac{\partial v_k}{\partial x_k} = \nabla \cdot \vec{v}$$

Recall that the divergence measures the rate of expansion/shrinking of a fluid element, which is how the fluid element deforms along the direction of velocity. The components which are off the diagonal thus represent the rate of shear deformation.

The antisymmetric part, on the other hand, is called the vorticity tensor. Notice how it looks similar to parts of the curl of the velocity. Naturally, the vorticity tensor is related to the rotation of the fluid.

However, we claim that the antisymmetric part must disappear, meaning the effects of viscosity are purely due to deformation of the fluid and not to due to rotation. To see this, let's again consider a counterexample.

Exercise 7.2 Consider a fluid under solid-body rotation, meaning the whole fluid rotates about the origin with angular velocity $\vec{\Omega}$, so the velocity field is given by $\vec{v} = \vec{\Omega} \times \vec{r}$ where $\vec{r}$ is the position given by $x_i \hat{\mathbf{x}}_i$.

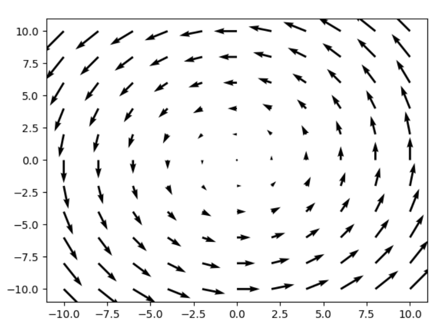


Figure 7.1: Solid Body Rotation

(a) Give a physical argument as to why the viscous stress tensor (along with any shear stress) should vanish in this case. (Hint: particles in solid objects don't have relative motion)

(b) Write the components of the velocity v_i using the Levi-Civita symbol. Then show that the symmetric part does disappear, while the antisymmetric part is nonzero under this type of motion, meaning it cannot appear in the viscous stress tensor. (Hint: Given the derivative $\partial x_i / \partial x_j$, what is it equal to when $i = j$ and $i \neq j$? Does this remind you of a particular tensor?)

We can now attach a tensor of proportionality to the velocity derivatives (a constant of proportionality for every combination of indices):

$$\sigma'_{ij} = A_{ijkl} \left. \frac{\partial v_k}{\partial x_l} \right|_{\text{sym}}$$

Before continuing with the derivation, let's ask: how constant is this tensor, or what other dependence can the viscous stress tensor have?

Earlier we argued the fluid should be isotropic, but we can also it should also not depend directly on position. The flow can depend on position, but the fluid itself should be the same everywhere we look, so the stress can only depend on position as far as the flow does. This assumption is what we call **homogeneity**. A fluid that is homogeneous is translation invariant, making it the counterpart to isotropy.

In the end, the only other things the viscous stress tensor can depend on are thermodynamic properties like temperature or density. However, the assumptions we have made thus far are enough to tell us what the viscous stress tensor should look like.

7.3 Symmetry of the Stress Tensor

There is one more fact we can use to simplify the form of the stress tensor, which is that it must be symmetric, or $\sigma_{ij} = \sigma_{ji}$. We can show that this is a result of Newton's 2nd Law for Rotations, but the derivation will be delayed until the end of the handout (see *Section 10.2*) since we have not developed all the required tools yet.

Instead, we will provide some intuition as to how the rotations are related to the symmetry of the stress tensor. Let us go back to the components of stress on a fluid elements:

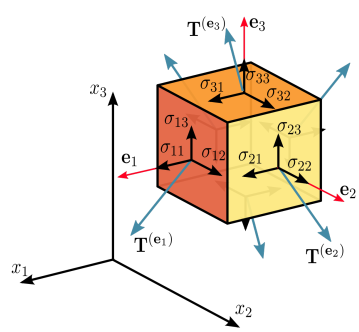


Figure 7.2: Components of stress

σ_{32} is a tangential stress which causes a clockwise torque about the $\hat{\mathbf{e}}_1$ axis while σ_{23} causes a counterclockwise torque about the same axis. We can see the same is true for the other axes too. If the stress tensor is symmetric, then there must be no net torque on the object.

To see why this should be true, let's consider Newton's 2nd Law for Rotations. For any arbitrary body fluid, total angular momentum should be conserved as long as there is no external torque. However, angular momentum is not conserved for individual fluid elements since tangential stresses will cause torque and exchange angular momentum between fluid elements.

Now just consider a single fluid element. Since there are no other fluid elements to exchange angular momentum with, it follows the net torque should be 0 in the absence of external torques. Thus, the stress tensor should be symmetric. This is still a handwavy explanation, but it will do for now.

7.4 Derivation of the Stress Tensor

Exercise 7.3

(a) Stating which assumptions you used, show that the rank-4 tensor A_{ijkl} can be rewritten in terms of two coefficients:

$$A_{ijkl} = \xi' \delta_{ij}\delta_{kl} + \eta(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$$

(b) Plug this back into the viscous stress tensor, and show that it must be given by

$$\sigma'_{ij} = \xi' \delta_{ij}\theta + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)$$

where $\theta = \nabla \cdot \vec{v}$. (Hint: Remember the Kronecker delta can be thought of as "setting" coefficients equal to each other.)

(c) As we argued earlier, viscosity resists deformation, which has two effects: tangential stresses which resist the sliding between fluid layers, and a resistance to deformation along the direction of velocity, which affects the normal stress.

Thus, we would like to separate the trace and traceless part of the viscous stress tensor. Do so by rewriting ξ' in terms of a new coefficient ξ using the substitution $\xi' = \xi - \frac{2}{3}\eta$. Collect the terms with ξ and η separately, and confirm that the part with η is traceless, while the part ξ has trace.

To summarize, we have shown the viscous stress is purely due to the deformation of the fluid, and is given by

$$\sigma'_{ij} = \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\theta \right) + \xi \delta_{ij}\theta$$

in terms of two coefficients η and ξ which should depend only on thermodynamic properties. η is called the dynamic viscosity, and ξ is called the bulk viscosity. So why did we separate the trace and traceless part?

Earlier we mentioned that the symmetric part of the velocity derivatives is called the rate of strain tensor, and we mentioned that the trace is equal to the divergence. We see here the bulk viscosity is responsible for this trace part, meaning it measures the resistance to change in volume (physical interpretation of divergence). Although we won't show this here, the traceless part is responsible for the change in shape of the fluid.

Thus, we split the deformation of the fluid into two parts: the change in volume of the fluid, and the change in shape.

Plugging this back into the stress tensor, we have

$$\sigma_{ij} = -p\delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\theta \right) + \xi \delta_{ij}\theta$$

We are actually now able to give a more accurate definition of pressure by including the effects of bulk viscosity. We define the mechanical pressure to be the average normal stress experienced by a fluid element (considering sign convention):

$$p_{\text{mech}} \equiv -\frac{1}{3}\sigma_{ii} = p - \xi\theta$$

However, we see that these two become the same thing when the fluid is incompressible, or $\theta = \nabla \cdot \vec{v} = 0$.

8 Viscous Equations of Motion

8.1 Cauchy Momentum Equation

We have one last question to answer: How do we include the stress tensor in our equations of motion?

It's quite simple, because we have already done this once. Let's once again consider the integral form of the momentum equation, over a region V enclosed by a surface S:

$$\int_V \rho \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right] dV = \int_V \vec{f}\, dV$$

We want to describe the net force on the region, now including the effects of viscosity.

Exercise 8.1 Similar to what we did for pressure, we will consider the force on the surface first, then convert it to a volume integral.

(a) Consider a small surface element dS with normal $\hat{n}$ which we can write in component form as $n_i\hat{\mathbf{x}}_i$. Using the stress tensor, write the net force on this surface element.

(b) Now write the net force as a surface integral, and convert it to a volume integral. (You might want to continue using indices here.)

(c) Thus derive the Cauchy Momentum Equation:

$$\frac{\partial v_i}{\partial t} + (\vec{v}\cdot\nabla)v_i = \frac{1}{\rho}\nabla_j\sigma_{ij}$$

or in vector form:

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = \frac{1}{\rho}\nabla\cdot\sigma$$

8.2 Navier-Stokes Equation

And well theres one last step:

Exercise 8.2

(a) Plug in the stress tensor to the Cauchy momentum equation. Earlier we said that η and ξ do depend on thermodynamic quantities, which could depend on position, but let us assume that η and ξ vary a negligible amount over position and are constant (which is true as long as those thermodynamic quantities do not vary greatly).

Thus derive the **Compressible Navier-Stokes Equation**:

$$\rho\left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right] = -\nabla p + \eta\nabla^2\vec{v} + \left(\xi + \frac{1}{3}\eta\right)\nabla\theta$$

(b) Assume incompressibility and define the kinematic viscosity $\nu = \eta/\rho$ to arrive at the **Incompressible Navier Stokes-Equation**:

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = -\frac{\nabla p}{\rho} + \nu\nabla^2\vec{v}$$

And there you have it! The Navier-Stokes Equation!

9 Closing Remarks

Believe it or not, you have just derived the Navier-Stokes equation, completely from scratch! It did get pretty complicated towards the end with all the indices everywhere, but hey, you made it through.

There are few technicalities about the derivation that I'd like to point out. When we derived the Cauchy momentum equation, we in fact implicitly assumed the continuity equation. This is not really a problem in most cases where we have a single fluid, but in the case where there are multiple fluids in which there is some exchange of mass by, for example, a chemical process, the Cauchy momentum equation would not hold.

There is a way to avoid this though, which is to write the momentum equation in conservation form, introducing a momentum flux tensor. If we had gone this way, we would have, at some point, had to explicitly use the continuity equation to obtain the Navier-Stokes. However, it was a bit convoluted, so I decided to go with the more straightforward derivation here.

The other thing is that we mostly avoided talking about thermodynamics. As we mentioned in the ideal section, we need an equation of state to make our system of equations closed, and especially in the viscous case, where there is exchange of energy through viscosity, thermodynamics can be very important.

Along with all the other assumptions that we made, it's clear that there are a lot of interesting things to be said in edge cases where those assumptions break down. It really was sort of an overstatement to say Navier-Stokes describes all of fluid dynamics. We haven't even touched on basic topics like surface tension, etc.

Nevertheless, I hope this handout has made you feel like you can step foot into the world of fluids and grab onto something, even if it's very small. Hopefully you can now start to see how these mathematical equations are motivated by physical ideas. There's a good reason why I avoided dumping half a chapter of a linear algebra textbook in favor of motivating tensors through what we physically needed.

So what's next? I don't know, maybe take a fluid dynamics class. Or maybe look into related subjects and see what you can dig up. Writing this handout has made me realize how vast fluid mechanics is as a subject, and continuum mechanics even more so - really a case of the Dunning-Kruger effect.

Anyways, thanks for reading and I hope you've had fun.

10 Appendix

10.1 Index Manipulation Practice

This section gives practice problems relating to the content of *Section 1.5* in preparation for *Section 7* **and assumes knowledge up to *Section 5*.**

10.1.1 Examples

Two useful vector calculus identities we will need are:

1. $\nabla \times \nabla f = 0$
2. $\nabla \cdot (\nabla \times \vec{F}) = 0$

There are intuitive geometric explanations for these identities, but our goal is to practice using index notation. Let us do the first one as an example

Using the Levi-Civita symbol we rewrite the curl:

$$(\nabla \times \nabla f)_i = \epsilon_{ijk} \nabla_j (\nabla f)_k$$

But the kth component of ∇f is precisely the partial of derivative of f respect to x_k, so we have

$$\epsilon_{ijk} \nabla_j (\nabla f)_k = \epsilon_{ijk} \nabla_j \nabla_k f$$

Note that we can exchange the order of derivatives. Here we can't do anything fancy with the Levi-Civita because we only have one. To show that this expression is 0, we need to directly invoke the properties of the Levi-Civita. Note that when $j = k$, the Levi-Civita symbol is 0. Note that when switching the indices j and k, the derivatives stay the same because order doesn't matter, but the Levi-Civita symbol flips sign since it is antisymmetric. Thus, we have

$$\epsilon_{ijk} \nabla_j \nabla_k f = -\epsilon_{ikj} \nabla_k \nabla_j f$$

So for $j \neq k$ the terms cancel out in pairs, and we have

$$(\nabla \times \nabla f)_i = \epsilon_{ijk} \nabla_j \nabla_k f = 0$$

thus showing $\nabla \times \nabla f = 0$.

The other thing we need to be able to work with is double cross products. They are unwieldy so we would like to be able to rewrite them in term of other operators. Let us show the identity

$$\nabla \times \nabla \times \vec{F} = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$$

Let's rewrite the left curl operator using Levi-Civita:

$$(\nabla \times \nabla \times \vec{F})_i = \epsilon_{ijk} \nabla_j (\nabla \times \vec{F})_k$$

Now we rewrite the second curl as well:

$$(\nabla \times \vec{F})_k = \epsilon_{klm} \nabla_l F_m$$

Note we don't reuse indices. This gives us

$$\epsilon_{ijk} \nabla_j (\nabla \times \vec{F})_k = \epsilon_{ijk} \epsilon_{klm} \nabla_j \nabla_l F_m$$

Before we rewrite the two Levi-Civitas as Kronecker deltas, we need the k to be the first index in both Levi-Civita symbols. To do this, we use the cyclic property of the Levi-Civita: $\epsilon_{ijk} = \epsilon_{kij}$. (The antisymmetric property also works, but you just end up with an extra negative at the end.) This gives us

$$\epsilon_{kij}\epsilon_{klm}\nabla_j\nabla_l F_m = (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\nabla_j\nabla_l F_m$$

Remember what the Kronecker delta does is set indices equal to each other, getting rid of one of them. Here we are looking for $(\nabla \times \nabla \times \vec{F})_i$, so we want to preserve the i index. We have

$$\begin{aligned}(\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\nabla_j\nabla_l F_m &= \nabla_j\nabla_i F_j - \nabla_j\nabla_j F_i \\ &= \nabla_i(\nabla_j F_j) - \nabla_j^2 F_i\end{aligned}$$

Finally, we can get rid of the j index by rewriting it in terms of vectors, giving

$$(\nabla \times \nabla \times \vec{F})_i = \nabla_i(\nabla_j F_j) - \nabla_j^2 F_i = \nabla_i(\nabla \cdot \vec{F}) - \nabla^2 F_i$$

Thus, we have

$$\nabla \times \nabla \times \vec{F} = \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$$

10.1.2 Exercises

These exercises are quite challenging (particularly the problems in the second one), but you will be more than prepared to tackle *Section 7* after this.

Exercise 10.1 In a similar fashion to the first example, show the identity $\nabla \cdot (\nabla \times \vec{F}) = 0$.

Exercise 10.2 These problems will give you plenty of practice with double cross products, and more importantly, dealing with lots of Kronecker deltas. In the spirit of the handout, let's use a physics example.

As briefly mentioned in *Section 1.4*, the curl of velocity is called the vorticity, which is a very important quantity in fluids. We write it as $\vec{\omega} = \nabla \times \vec{v}$. Naturally we are interested in how it changes over time, or $d\vec{\omega}/dt$.

We start with the Euler equation and take the curl of both sides, giving

$$\frac{\partial \vec{\omega}}{\partial t} + \nabla \times [(\vec{v} \cdot \nabla)\vec{v}] = 0$$

(a) Show that we can write

$$(\vec{v} \cdot \nabla)\vec{v} = \frac{1}{2}\nabla v^2 - \vec{v} \times \vec{\omega}$$

(Hint: Start by expanding $\vec{v} \times \vec{\omega}$, which is actually a double cross product. The $\frac{1}{2}\nabla v^2$ comes from an application of reverse product rule.)

(b) Thus show we have

$$\frac{\partial \vec{\omega}}{\partial t} = \nabla \times (\vec{v} \times \vec{\omega})$$

(c) Again expand the double cross product to show

$$\nabla \times (\vec{v} \times \vec{\omega}) = (\vec{\omega} \cdot \nabla)\vec{v} - (\vec{v} \cdot \nabla)\vec{\omega} - \vec{\omega}(\nabla \cdot \vec{v})$$

(Hint: the result you showed in *Exercise 10.1* will now come in handy.)

(d) Finally show that

$$\frac{d\vec{\omega}}{dt} = (\vec{\omega} \cdot \nabla)\vec{v} - \vec{\omega}(\nabla \cdot \vec{v})$$

10.2 Derivation of the Symmetry of the Stress Tensor

This section assumes you have read through *Section 8.*

Now that we have the Cauchy Momentum equation, this simply becomes a calculation problem. We will start with Newton's 2nd Law for Rotation and derive the symmetry of the stress tensor.

Exercise 10.3 Newton's 2nd Law for Rotation says $\vec{\tau} = d\vec{L}/dt$ where $\vec{\tau}$ is the torque and $\vec{L}$ is the angular momentum. We have the following definitions: $\vec{\tau} = \vec{r} \times \vec{F}$ and $\vec{L} = \vec{r} \times m\vec{v}$, where $\vec{r}$ is the displacement from the axis of rotation.

(a) Show that

$$\oint_S \vec{r} \times \sigma\hat{n}\, dS = \frac{d}{dt}\int_V \vec{r} \times \vec{v}\rho\, dV$$

Remember Newton's 2nd Law holds for particles so we follow a region with fixed mass over time.

(b) We can rewrite this in index form as

$$\oint_S \epsilon_{ijk} x_j \sigma_{kl} n_l\, dS = \frac{d}{dt}\int_V \epsilon_{ijk} x_j v_k \rho\, dV$$

Show the LHS can be rewritten as

$$\int_V \epsilon_{ijk}\left(\sigma_{kj} + x_j \frac{\partial \sigma_{kl}}{\partial x_l}\right) dV$$

Also show the RHS can be rewritten as

$$\int_V \epsilon_{ijk} x_j \frac{dv_k}{dt}\rho\, dV$$

(c) Show that this implies

$$\epsilon_{ijk}\sigma_{kj} = 0$$

(Hint: Apply the Cauchy Momentum equation.)

(d) We can avoid directly dealing with the Levi-Civita by multiplying both sides by another Levi-Civita ϵ_{imn}. Use this to show that the stress tensor must be symmetric.

Now we have finally closed every thread we promised to close.

11 References

[1] Scoccimaro, R. (2024). Introduction to Fluid Dynamics Lecture Notes

[2] Ali-Haimoud, Y. (2023). Mathematical Physics. NYU Center for Cosmology and Particle Physics. https://cosmo.nyu.edu/yacine/teaching/MathPhys.pdf

[3] Kundu, P. K., Cohen, I. M., & Dowling, D. R. (2016). Fluid Mechanics (6th ed.). Academic Press.

[4] Coleman, N. (2010). A derivation of the navier-stokes equations. Ball State University. https://digitalresearch.bsu.edu/mathexchange/wp-content/uploads/2021/02/A-Derivation-of-the-Navier-Stokes-Equations_coleman.neal_.pdf

[5] van den Bosch, F. (n.d.). Chapter 6 viscosity & the stress tensor. Yale University. http://www.astro.yale.edu/vdbosch/astro320_summary6.pdf

[6] Chua, D. (2015). Invariant and isotropic tensors. IA Vector Calculus. https://dec41.user.srcf.net/h/IA_L/vector_calculus/15_1

[7] Treil, S. (2021, January 11). Linear Algebra Done Wrong. Brown University. https://www.math.brown.edu/streil/papers/LADW/LADW_2021_01-11.pdf

[8] Bower, A. F. (2012) Continuum Mechanics - Conservation Laws. Brown University. https://www.brown.edu/Departments/Engineering/Courses/En221/Notes/Conservation_Laws/Conservation_Laws.htm

Solutions to:

From Newton to Navier-Stokes

Section 3

3.1)

(a) $\vec{x}(t) = \vec{x}_0 + \vec{\psi}(\vec{x}_0, t)$

(b) Differentiating gives

$$\vec{\mathrm{v}}(t) = \frac{d\vec{x}}{dt} = \frac{d\vec{\psi}}{dt}(\vec{x}_0, t)$$

3.2) In order for both descriptions to be equivalent, the instantaneous velocity at a given position and a given time should agree. However, this means the position we plug into the Eulerian velocity field also depends on time, since the position of a given fluid element depends on time. Thus, we should have

$$\vec{\mathrm{v}}(t) = \vec{v}\,[\vec{x}(t), t]$$

Then, using the results from the previous exercise, we get

$$\frac{d\vec{\psi}}{dt}(\vec{x}_0, t) = \vec{v}\left[\vec{x}_0 + \vec{\psi}(\vec{x}_0, t), t\right]$$

Section 4

4.1) Here we track a specific fluid element over time, so the position also depends on time. Thus, chain rule gives

$$\frac{d\phi}{dt} = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial x}\frac{dx}{dt} + \frac{\partial\phi}{\partial y}\frac{dy}{dt} + \frac{\partial\phi}{\partial z}\frac{dz}{dt} = \frac{\partial\phi}{\partial t} + \nabla\phi\cdot\vec{v} = \frac{\partial\phi}{\partial t} + (\vec{v}\cdot\nabla)\phi$$

Using index notation gives a slightly more compact way of writing chain rule:

$$\frac{d\phi}{dt} = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial x_i}\frac{dx_i}{dt} = \frac{\partial\phi}{\partial t} + \left(\frac{dx_i}{dt}\frac{\partial}{\partial x_i}\right)\phi = (\vec{v}\cdot\nabla)\phi$$

We write $\vec{v}\cdot\nabla$ to avoid ambiguity, which will become clear when we start dealing with vector fields.

When we take a partial derivative w.r.t. t, we keep the spatial coordinates x, y, z fixed, but we have to remember the fluid element itself also moves over time, so after a small time δt, the fluid element at the original position may not be the same one. Thus, the term with the gradient dotted into the velocity accounts for this change in position.

4.2)

(a) From *Exercise 3.2*, we have

$$\frac{d}{dt}\vec{\mathrm{v}}(t) = \frac{d}{dt}\vec{v}\,[\vec{x}(t), t] = \frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}$$

(b) Let us separate out the components of position, so

$$\vec{v}\,[\vec{x} + \delta\vec{x}, t + \delta t] = \vec{v}\,[x + \delta x, y + \delta y, z + \delta z, t + \delta t]$$

where $\delta\vec{x} = \vec{v}\,\delta t = (v_x\,\delta t)\hat{\mathbf{x}} + (v_y\,\delta t)\hat{\mathbf{y}} + (v_z\,\delta t)\hat{\mathbf{z}}$

Doing a first-order Taylor expansion gives

$$\begin{aligned}
\vec{v}\,[x+\delta x, y+\delta y, z+\delta z, t+\delta t] &\approx \vec{v}\,[x,y,z,t] + \frac{\partial \vec{v}}{\partial x}\delta x + \frac{\partial \vec{v}}{\partial y}\delta y + \frac{\partial \vec{v}}{\partial z}\delta z + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \vec{v}\,[x,y,z,t] + \frac{\partial \vec{v}}{\partial x}v_x\,\delta t + \frac{\partial \vec{v}}{\partial y}v_y\,\delta t + \frac{\partial \vec{v}}{\partial z}v_z\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \vec{v}\,[x,y,z,t] + \left(v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)\vec{v}\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \vec{v}\,[x,y,z,t] + (\vec{v}\cdot\nabla)\vec{v}\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t
\end{aligned}$$

Then,

$$\delta\vec{v} \approx \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right]\delta t$$

or, to be more precise,

$$\delta\vec{v} = \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right]\delta t + \mathcal{O}[(\delta t)^2]$$

Thus, we have

$$\frac{d\vec{v}}{dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}$$

Alternatively, here is the same derivation using index notation:

$$\begin{aligned}
\delta\vec{v} &\approx \frac{\partial \vec{v}}{\partial x_i}\delta x_i + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \frac{\partial \vec{v}}{\partial x_i}v_i\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \left(v_i\frac{\partial}{\partial x_i}\right)\vec{v}\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right]\delta t
\end{aligned}$$

Note: Here it becomes clear why we choose to write $\vec{v}\cdot\nabla$. It is a bit trickier to define the gradient of a vector field, but since we are dotting it with the velocity anyways, we can think of $\vec{v}\cdot\nabla$ as one single operator given by

$$\vec{v}\cdot\nabla = v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z} = v_i\frac{\partial}{\partial x_i}$$

4.3

(a) A velocity normal to the surface causes a change in volume, so for an infinitesimal surface element $d\vec{S}$, the rate of change in volume is given by $\vec{v}\cdot d\vec{S}$. Integrating, the net rate of change in volume is given by

$$\oint_S \vec{v}\cdot d\vec{S}$$

(b) This should be equal to the time derivative of the volume, so we have

$$\frac{d}{dt}\int_V dV = \oint_S \vec{v}\cdot d\vec{S}$$

Applying the Divergence Theorem, we have

$$\frac{d}{dt}\int_V dV = \int_V \nabla\cdot\vec{v}\,dV$$

(c) Taking an infinitesimal volume element δV, we can get rid of the integrals, giving

$$\frac{d}{dt}(\delta V) = \nabla \cdot \vec{v}\, \delta V$$

Thus,

$$\nabla \cdot \vec{v} = \frac{1}{\delta V}\frac{d\delta V}{dt}$$

4.4

(a) By product rule, and then *Exercise 4.3(c)*, we have

$$\frac{d}{dt}(\xi\, \delta V) = \xi\frac{d\delta V}{dt} + \delta V\frac{d\xi}{dt} = \xi\nabla \cdot \vec{v}\, \delta V + \delta V\frac{d\xi}{dt} = \left(\xi\nabla \cdot \vec{v} + \frac{d\xi}{dt}\right)\delta V$$

Here we see the product rule has a physical interpretation as well. We not only have to account for the change in ξ as the particle moves (given by the material derivative), but since it is a unit-volume quantity, we also have to account for the expansion/shrinking of the fluid element (given by the divergence term).

(b) Integrating over an arbitrary volume, we get

$$\frac{d}{dt}\int_V \xi\, dV = \int_V \left(\xi\nabla \cdot \vec{v} + \frac{d\xi}{dt}\right) dV$$

4.5

(a) Again by product rule, we have

$$\frac{d}{dt}(\varphi\rho\, \delta V) = \rho\, \delta V\frac{d\varphi}{dt} + \varphi\frac{d}{dt}(\rho\, \delta V)$$

However, $\rho\, \delta V$ is the mass of the infinitesimal fluid element which we are following. Remember we consider fluid elements with fixed mass, so it may expand or shrink over time, but the mass will not change. Thus, the time derivative disappears and we have

$$\frac{d}{dt}(\varphi\rho\, \delta V) = \rho\, \delta V\frac{d\varphi}{dt}$$

Unlike the previous exercise, a fixed mass means we only have to account for the particle motion with the material derivative.

(b) Again, integrating over an arbitrary volume, we get

$$\frac{d}{dt}\int_V \varphi\rho\, dV = \int_V \rho\frac{d\varphi}{dt}\, dV$$

Section 5

5.1

(a) $m = \displaystyle\int_V \rho\, dV$

(b) From *Exercise 4.4*, we have

$$\frac{dm}{dt} = \frac{d}{dt}\int_V \rho\, dV = \int_V \left(\rho\nabla\cdot\vec{v} + \frac{d\rho}{dt}\right) dV = 0$$

Since this holds for any arbitrary volume, we can take an infinitesimal volume, getting rid of the integral, giving

$$\rho\nabla\cdot\vec{v} + \frac{d\rho}{dt} = 0$$

Here we expand the material derivative, giving

$$\frac{\partial\rho}{\partial t} + (\vec{v}\cdot\nabla)\rho + \rho\nabla\cdot\vec{v} = 0$$

We can see that if we show $\nabla\cdot(\rho\vec{v}) = (\vec{v}\cdot\nabla)\rho + \rho\nabla\cdot\vec{v}$, then we're done. Let us show this using index notation:

$$\begin{aligned}\nabla\cdot(\rho\vec{v}) &= \nabla_i(\rho v_i)\\ &= \rho\nabla_i v_i + v_i\nabla_i\rho\\ &= \rho\nabla\cdot\vec{v} + (\vec{v}\cdot\nabla)\rho\end{aligned}$$

Thus, we have the continuity equation

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0$$

(c) Assuming incompressibility ($\rho = \text{const.}$), we have $\nabla\cdot\vec{v} = 0$ (a.k.a the incompressibility condition).

5.2

(a) The mass flux is given by

$$\oint_S \rho\vec{v}\cdot d\vec{S}$$

(b) The time derivative of the mass should be the same as the mass flux. However, since a positive flux is a decrease in mass, we need to add a negative sign. Thus, we have

$$\frac{\partial}{\partial t}\int_V \rho\, dV = -\oint_S \rho\vec{v}\cdot d\vec{S}$$

Note that we use a partial derivative to clarify that we are working in the Eulerian perspective. Then, applying the divergence theorem and moving everything to one side gives us

$$\frac{\partial}{\partial t}\int_V \rho\, dV + \int_V \nabla\cdot(\rho\vec{v})\, dV = 0$$

(c) Since we are working with a fixed region, the spatial coordinates when we integrate aren't time dependent, so we can directly move the derivative inside the integral. Then, taking an infinitesimal volume element, we again get the differential form of the continuity equation:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot(\rho\vec{v}) = 0$$

5.3

(a) $\vec{p} = \int_V \vec{v}\rho\, dV$

(b) From *Exercise 4.5*, we have

$$\int_V \vec{f}\, dV = \vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}\int_V \vec{v}\rho\, dV = \int_V \rho\frac{d\vec{v}}{dt}\, dV$$

Taking an infinitesimal volume, we have

$$\rho\frac{d\vec{v}}{dt} = \vec{f}$$

Then, expanding the material derivative and dividing by ρ gives the momentum equation:

$$\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = \frac{\vec{f}}{\rho}$$

5.4

(a) Let us use index notation for readability. We have $\nabla p = \hat{\mathbf{x}}_i\nabla_i p$ and we can write $\nabla_i p = \nabla\cdot(p\hat{\mathbf{x}}_i)$. Thus,

$$\int_V \nabla p\, dV = \hat{\mathbf{x}}_i\int_V \nabla\cdot(p\hat{\mathbf{x}}_i)\, dV$$

Then, by divergence theorem,

$$\hat{\mathbf{x}}_i\int_V \nabla\cdot(p\hat{\mathbf{x}}_i)\, dV = \hat{\mathbf{x}}_i\oint_S p\hat{\mathbf{x}}_i\cdot d\vec{S} = \oint_S p(\hat{\mathbf{x}}_i\cdot d\vec{S})\hat{\mathbf{x}}_i = \oint_S p\, d\vec{S}$$

(b) Thus, we have

$$\int_V \vec{f}\, dV = \vec{F} = -\int_V \nabla p\, dV$$

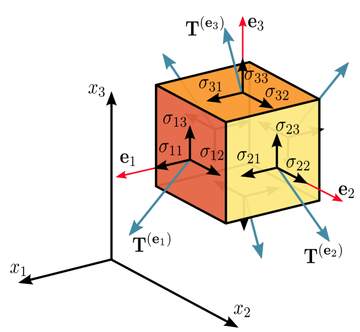
Since this is true for any arbitrary volume, we must have $\vec{f} = -\nabla p$ (this is ignoring long-range forces, which we can always add back later), and we can modify the momentum equation to get the Euler equation:

$$\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = -\frac{\nabla p}{\rho}$$

Section 6

6.1

(a) σ_{ij} can be interpreted as the stress in the x_i through a surface with normal in the x_j. The normal stresses are σ_{11}, σ_{22}, and σ_{33}, and the other entries are the tangential stresses. This is why you often see this diagram for the stress tensor: (Credit: Wikipedia)



(b) Using multilinearity, we have

$$\sigma(\vec{v},\vec{w}) = v_1\sigma(\hat{\mathbf{x}}_1,\vec{w}) + v_2\sigma(\hat{\mathbf{x}}_2,\vec{w}) + v_3\sigma(\hat{\mathbf{x}}_3,\vec{w}) = v_i\sigma(\hat{\mathbf{x}}_i,\vec{w})$$

Then, we can do the same thing to expand $\vec{w}$, giving

$$\sigma(\vec{v},\vec{w}) = v_i\sigma(\hat{\mathbf{x}}_i,\hat{\mathbf{x}}_j)w_j = v_i\sigma_{ij}w_j$$

(c) i. We can rewrite $\sigma_{ij}w_j$ as $\sigma\vec{w}$ using the rules of matrix-vector multiplication (remember j is being summed over), which collects $\sigma_{ij}w_j$ for each i as the components of the resulting vector:

$$\sigma\vec{w} = \begin{bmatrix}\sigma_{11} & \sigma_{12} & \sigma_{13}\\ \sigma_{21} & \sigma_{22} & \sigma_{23}\\ \sigma_{31} & \sigma_{32} & \sigma_{33}\end{bmatrix}\begin{bmatrix}w_1\\ w_2\\ w_3\end{bmatrix} = \begin{bmatrix}\sigma_{11}w_1+\sigma_{12}w_2+\sigma_{13}w_3\\ \sigma_{21}w_1+\sigma_{22}w_2+\sigma_{23}w_3\\ \sigma_{31}w_1+\sigma_{32}w_2+\sigma_{33}w_3\end{bmatrix} = \begin{bmatrix}\sigma_{1j}w_j\\ \sigma_{2j}w_j\\ \sigma_{3j}w_j\end{bmatrix}$$

So $(\sigma\vec{w})_i = \sigma_{ij}w_j$.

ii. From part (a), we can interpret $1\cdot\sigma_{ij}\cdot w_j$ as the stress in the x_i direction through a surface with normal $\vec{w}$ (again j is summed over). Thus $\sigma\vec{w}$ can just be interpreted as the net stress vector (force per unit area) through a surface with normal $\vec{w}$. This means we can also interpret the stress tensor as a linear transformation $\sigma : V \to V$ that takes in the normal to the surface, and gives a stress vector back.

iii. We have $v_iu_i = \vec{v}\cdot\vec{u}$. Since $\vec{u}$ is the net stress, if we want the stress in a specific direction $\vec{v}$ (let's assume it's a unit vector), we can just project $\vec{u}$ onto the direction by taking a dot product.

6.2

(a) i. We have $T_{23} = R_{2p}R_{3q}T_{pq} = R_{21}R_{33}T_{13} = T_{13}$. Thus $T_{13} = -T_{13} = 0$ and also $T_{23} = 0$.

ii. We have $T_{31} = R_{33}R_{12}T_{32} = -T_{32}$ and $T_{32} = R_{33}R_{21}T_{31} = T_{31}$. Thus, $T_{31} = T_{32} = 0$.

iii. We have $T_{11} = R_{12}R_{12}T_{22} = T_{22}$.

(b) i. We have $T_{12} = R_{11}R_{23}T_{13} = -T_{13} = 0$. Similarly, $T_{21} = R_{23}R_{11}T_{31} = -T_{31} = 0$.

ii. We have $T_{22} = R_{23}R_{23}T_{33} = T_{33}$.

(c) Thus, all the entries not on the diagonal are 0 and we have $T_{11} = T_{22} = T_{33}$. By letting $T_{11} = \alpha$, we have

$$T_{ij} = \alpha\delta_{ij}$$

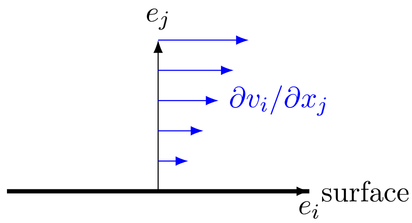
as desired.

Section 7

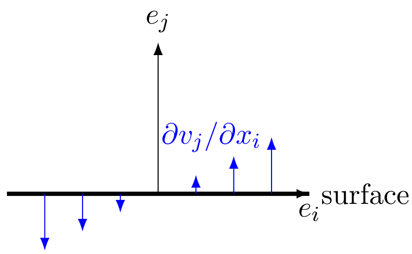
7.1 Since σ_{ij} represents the stress in the x_i direction a surface with x_j normal, naturally we might expect the derivatives to have indices of i and j, but let's make this intuition a bit more concrete.

For normal stresses (e.g. σ_{ii} where i is not summed over), part of it is due to pressure as we explained, but the other part it should be due to stretching along the direction of motion, meaning it should depend on derivatives with the same indices, like $\partial v_i/\partial x_i$, and it wouldn't make too much sense for it to depend on the other derivatives.

Recall the diagram of shear stress from *Section 2.1.* For shear stresses we expect two contributions. The first one is $\partial v_i/\partial x_j$, which represents velocity field in the x_i direction which varies along the x_j direction. This describes the sliding between fluid layers that we identify with friction in solids.



The second contribution is $\partial v_j/\partial x_i$, which describes velocity normal to the surface that vary along the surface. This tends to rotate the surface, and as we will see, viscosity is related to the deformation of the fluid.



7.2

(a) Consider a solid disk spinning with angular velocity $\vec{\Omega}$. The instantaneous velocity of a point on the disk at position $\vec{r}$ away from the center would also be given $\vec{\Omega} \times \vec{r}$. However, in the case of a solid disk, we think of there being no relative movement between particles, and we think of the disk moving together as a whole. Thus, intuitively, any shear stress should disappear in this case.

(b) We have $v_i = \epsilon_{imn}\Omega_m x_n$. Plugging this into the symmetric part, we get

$$\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} = \frac{1}{2}\left(\frac{\partial}{\partial x_l}\epsilon_{kmn}\Omega_m x_n + \frac{\partial}{\partial x_k}\epsilon_{lmn}\Omega_m x_n\right)$$

The derivative only affects x_n and we see that $\partial x_l/\partial x_n = 1$ when $l = n$ and 0 otherwise, which is exactly the Kronecker delta. Thus, we have

$$\begin{aligned}\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} &= \frac{1}{2}(\epsilon_{kmn}\Omega_m\delta_{ln} + \epsilon_{lmn}\Omega_m\delta_{kn}) \\ &= \frac{1}{2}\Omega_m(\epsilon_{kml} + \epsilon_{lmk}) \\ &= \frac{1}{2}\Omega_m(\epsilon_{kml} - \epsilon_{kml}) \\ &= 0\end{aligned}$$

Here we used the antisymmetric property of the Levi-Civita where swapping any two indices causes the sign to reverse. We see that the antisymmetric part just changes the sign in the middle, so we must have

$$\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{antisym}} = \frac{1}{2}\Omega_m(\epsilon_{kml} + \epsilon_{kml}) \neq 0$$

Since we argued that the viscous stress tensor should disappear under this motion, the antisymmetric part of cannot appear in it.

7.3

(a) Assuming the fluid is isotropic, we can write the rank-4 tensor as

$$A_{ijkl} = \xi' \delta_{ij}\delta_{kl} + \eta\delta_{ik}\delta_{jl} + \mu\delta_{il}\delta_{jk}$$

However, since we argued that the stress tensor is symmetric, meaning $\sigma_{ij} = \sigma_{ji}$, we must have $A_{ijkl} = A_{jikl}$, and we must have $\eta = \mu$. We keep η, leaving us with only two coefficients:

$$A_{ijkl} = \xi' \delta_{ij}\delta_{kl} + \eta(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$$

In fact it is possible to show the stress tensor is symmetric in general ($\sigma_{ij} = \sigma_{ji}$), but we already have a symmetry here that we can use.

(b) Plugging this back into the viscous stress tensor, we have

$$\begin{aligned}
\sigma'_{ij} &= \left[\xi'\delta_{ij}\delta_{kl} + \eta(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\right] \frac{1}{2}\left(\frac{\partial v_k}{\partial x_l} + \frac{\partial v_l}{\partial x_k}\right) \\
&= \xi'\delta_{ij}\frac{\partial v_k}{\partial x_k} + \eta\left[\frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j}\right)\right] \\
&= \xi'\delta_{ij}\theta + \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)
\end{aligned}$$

(c) Substituting $\xi' = \xi - \frac{2}{3}\eta$, we have

$$\begin{aligned}
\sigma'_{ij} &= \left(\xi - \frac{2}{3}\eta\right)\delta_{ij}\theta + \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) \\
&= \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\theta\right) + \xi\delta_{ij}\theta
\end{aligned}$$

The trace is just given by σ'_{ii} (remember i is summed over):

$$\begin{aligned}
\operatorname{tr}\sigma'_{ij} &= \sigma'_{ii} \\
&= \eta\left(\frac{\partial v_i}{\partial x_i} + \frac{\partial v_i}{\partial x_i} - \frac{2}{3}\delta_{ii}\theta\right) + \xi\delta_{ii}\theta \\
&= \eta\left(\theta + \theta - \frac{2}{3}\cdot 3\theta\right) + \xi\cdot 3\theta \\
&= 0 + 3\xi\theta
\end{aligned}$$

As expected, the part with η is traceless, while the part with ξ contains the trace.

Section 8

8.1

(a) As we argued earlier, $\sigma\hat{n}$ or $\sigma_{ij}n_j$ gives the stress vector. Multiplying this by the area dS gives the force on that surface element $\sigma_{ij}n_j\,dS$ or $\sigma\cdot d\vec{S}$.

(b) Thus, the net force on the surface S is given by

$$F_i = \oint_S \sigma_{ij} n_j \, dS = \int_V \nabla_j \sigma_{ij} \, dV$$

or in vector form,

$$\vec{F} = \int_V \nabla \cdot \sigma \, dV$$

(c) Since

$$\int_V \vec{f} \, dV = \vec{F} = \int_V \nabla \cdot \sigma \, dV$$

for any volume V, we have $\vec{f} = \nabla \cdot \sigma$. Plugging this into the general momentum equation, we get

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = \frac{1}{\rho} \nabla \cdot \sigma$$

or

$$\frac{\partial v_i}{\partial t} + (\vec{v} \cdot \nabla) v_i = \frac{1}{\rho} \nabla_j \sigma_{ij}$$

which makes it clearer how we take the "divergence" of a rank-2 tensor.

8.2

(a) Taking the divergence of the stress tensor and assuming η and ξ to be constant, we have

$$\begin{aligned}
\nabla_j \sigma_{ij} &= \frac{\partial}{\partial x_j} \left[-p\delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_i}{\partial x_j} - \frac{2}{3} \delta_{ij} \theta \right) + \xi \delta_{ij} \theta \right] \\
&= -\frac{\partial p}{\partial x_i} + \eta \left(\frac{\partial^2 v_i}{\partial x_j^2} + \frac{\partial}{\partial x_i} \frac{\partial v_j}{\partial x_j} - \frac{2}{3} \frac{\partial \theta}{\partial x_i} \right) + \xi \frac{\partial \theta}{\partial x_i} \\
&= -\frac{\partial p}{\partial x_i} + \eta \left(\nabla^2 v_i + \frac{\partial \theta}{\partial x_i} - \frac{2}{3} \frac{\partial \theta}{\partial x_i} \right) + \xi \frac{\partial \theta}{\partial x_i} \\
&= -\frac{\partial p}{\partial x_i} + \eta \nabla^2 v_i + \left(\xi + \frac{1}{3} \eta \right) \frac{\partial \theta}{\partial x_i}
\end{aligned}$$

Thus we have

$$\nabla \cdot \sigma = -\nabla p + \eta \nabla^2 \vec{v} + \left(\xi + \frac{1}{3} \eta \right) \nabla \theta$$

Plugging this back into the Cauchy momentum equation and moving ρ to the LHS, we get:

$$\rho \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} \right] = -\nabla p + \eta \nabla^2 \vec{v} + \left(\xi + \frac{1}{3} \eta \right) \nabla \theta$$

which is the **Compressible Navier-Stokes Equation**.

(b) Assuming incompressibility (e.g. $\nabla \cdot \vec{v} = \theta = 0$) and defining $\nu = \eta/\rho$, we have

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = -\frac{\nabla p}{\rho} + \nu \nabla^2 \vec{v}$$

which is the **Incompressible Navier-Stokes Equation**!!!

Section 10

As there are already multiple examples in *Section 10.1*, and the value for the problems in this section comes from working through them yourself, the solutions will not go into as much detail.

10.1 We have

$$\begin{aligned}\nabla \cdot (\nabla \times \vec{F}) &= \nabla_i (\nabla \times \vec{F})_i \\ &= \nabla_i \epsilon_{ijk} \nabla_j F_k\end{aligned}$$

Again we will be making use of the antisymmetry of the Levi-Civita. Noting that the derivatives can be exchanged we have

$$\begin{aligned}\nabla \cdot (\nabla \times \vec{F}) &= \nabla_i \epsilon_{ijk} \nabla_j F_k \\ &= \nabla_j \epsilon_{ijk} \nabla_i F_k \\ &= -\nabla_j \epsilon_{jik} \nabla_i F_k \\ &= -\nabla_j (\nabla \times \vec{F})_j \\ &= -\nabla \cdot (\nabla \times \vec{F})\end{aligned}$$

Thus we must have

$$\nabla \cdot (\nabla \times \vec{F}) = 0$$

10.2

(a) Expanding $\vec{v} \times \vec{\omega}$, we have

$$\begin{aligned}(\vec{v} \times \vec{\omega})_i &= [\vec{v} \times (\nabla \times \vec{v})]_i \\ &= \epsilon_{ijk} \epsilon_{klm} v_j \nabla_l v_m \\ &= \epsilon_{kij} \epsilon_{klm} v_j \nabla_l v_m \\ &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) v_j \nabla_l v_m \\ &= v_j \nabla_i v_j - v_j \nabla_j v_i\end{aligned}$$

Note that product rule gives

$$\frac{1}{2} \nabla_i v_j^2 = v_j \nabla_i v_j$$

This gives

$$\begin{aligned}(\vec{v} \times \vec{\omega})_i &= \frac{1}{2} \nabla_i v_j^2 - v_j \nabla_j v_i \\ &= \frac{1}{2} \nabla_i v^2 - (\vec{v} \cdot \nabla) v_i\end{aligned}$$

Thus, we have

$$\vec{v} \times \vec{\omega} = \frac{1}{2} \nabla v^2 - (\vec{v} \cdot \nabla) \vec{v}$$

(b) Substituting this back into the "curled" Euler equation, we get

$$\frac{\partial \vec{\omega}}{\partial t} = \nabla \times (\vec{v} \times \vec{\omega})$$

(c) We again expand the double cross product, giving

$$\begin{aligned}
[\nabla \times (\vec{v} \times \vec{\omega})]_i &= \epsilon_{ijk}\epsilon_{klm}\nabla_j(v_l\omega_m) \\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\nabla_j(v_l\omega_m) \\
&= \nabla_j(v_i\omega_j) - \nabla_j(v_j\omega_i)
\end{aligned}$$

Note the parentheses. We will now expand the expression using product rule, and we will also use $\nabla \cdot \vec{\omega} = \nabla \cdot (\nabla \times \vec{v}) = 0$:

$$\begin{aligned}
\nabla_j(v_i\omega_j) - \nabla_j(v_j\omega_i) &= v_i\nabla_j\omega_j + \omega_j\nabla_j v_i - v_j\nabla_j\omega_i - \omega_i\nabla_j v_j \\
&= v_i\nabla \cdot \vec{\omega} + (\vec{\omega} \cdot \nabla)v_i - (\vec{v} \cdot \nabla)\omega_i - \omega_i\nabla \cdot \vec{v} \\
&= (\vec{\omega} \cdot \nabla)v_i - (\vec{v} \cdot \nabla)\omega_i - \omega_i\nabla \cdot \vec{v}
\end{aligned}$$

Thus, we have

$$\nabla \times (\vec{v} \times \vec{\omega}) = (\vec{\omega} \cdot \nabla)\vec{v} - (\vec{v} \cdot \nabla)\vec{\omega} - \vec{\omega}(\nabla \cdot \vec{v})$$

(d) Plugging this back into the equation from part (b), we have

$$\frac{\partial \vec{\omega}}{\partial t} = (\vec{\omega} \cdot \nabla)\vec{v} - (\vec{v} \cdot \nabla)\vec{\omega} - \vec{\omega}(\nabla \cdot \vec{v})$$

Moving the $(\vec{v} \cdot \nabla)\vec{\omega}$ term to the LHS and using the definition of the material derivative, we have

$$\frac{d\vec{\omega}}{dt} = (\vec{\omega} \cdot \nabla)\vec{v} - \vec{\omega}(\nabla \cdot \vec{v})$$

Exercise 10.3

(a) Recall in the derivation of the Cauchy Momentum equation that we wrote the force as

$$\oint_S \sigma\hat{n}\, dS$$

We can similarly write the torque as

$$\oint_S \vec{r} \times \sigma\hat{n}\, dS$$

We can also write the angular momentum as

$$\int_V \vec{r} \times \vec{v}\rho\, dV$$

Then, Newton's 2nd Law gives

$$\oint_S \vec{r} \times \sigma\hat{n}\, dS = \frac{d}{dt}\int_V \vec{r} \times \vec{v}\rho\, dV$$

(b) For the LHS, we apply the divergence theorem, giving

$$\begin{aligned}
\oint_S \epsilon_{ijk}x_j\sigma kln_l\, dS &= \int_V \frac{\partial}{\partial x_l}(\epsilon_{ijk}x_j\sigma_{kl})dV \\
&= \int_V \epsilon_{ijk}\left(\delta_{jl}\sigma_{kl} + x_j\frac{\partial\sigma_{kl}}{\partial x_l}\right)dV \\
&= \int_V \epsilon_{ijk}\left(\sigma_{kj} + x_j\frac{\partial\sigma_{kl}}{\partial x_l}\right)dV
\end{aligned}$$

For the RHS, since we follow the fluid region, we can use the results of *Exercise 4.5* to show that

$$\frac{d}{dt}\int_V \epsilon_{ijk}x_jv_k\rho\, dV = \int_V \epsilon_{ijk}x_j\frac{dv_k}{dt}\rho\, dV$$

(c) Setting the two sides equal to each other, we have

$$\int_V \epsilon_{ijk} \left(\sigma_{kj} + x_j \frac{\partial \sigma_{kl}}{\partial x_l} \right) dV = \int_V \epsilon_{ijk} x_j \frac{dv_k}{dt} \rho \, dV$$

Moving the term with the derivative of the stress tensor the RHS, we have

$$\int_V \epsilon_{ijk} \sigma_{kj} \, dV = \int_V \epsilon_{ijk} x_j \left(\rho \frac{dv_k}{dt} - \frac{\partial \sigma_{kl}}{\partial x_l} \right) dV$$

However, the expression in the parentheses on the RHS must be 0 because it exactly matches the terms in the Cauchy Momentum equation. This equation is also true for any arbitrary portion of fluid, meaning we can get rid of the integral. This leaves us with

$$\epsilon_{ijk} \sigma_{kj} = 0$$

(d) Multiplying both sides by ϵ_{imn}, we have

$$\begin{aligned} \epsilon_{imn} \epsilon_{ijk} \sigma_{kj} &= 0 \\ (\delta_{mj} \delta_{nk} - \delta_{mk} \delta_{nj}) \sigma_{kj} &= 0 \\ \sigma_{nm} - \sigma_{mn} &= 0 \end{aligned}$$

Finally, we have

$$\sigma_{mn} = \sigma_{nm}$$

as desired.